Supplementary Material for "Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures"

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S1. E-ACGL extension and path sensitivity

The main article defines E-CGL as the primary estimator and E-ACGL as its adaptive extension. This Supplementary Material gives the complete E-ACGL path specification, audit, and numerical results. Notation and abbreviations not redefined here follow the main article. Let E^{D} denote the common dense pilot used in the main article. For completeness, with $c_{\cdot j} = H_K E_{\cdot j}$, its penalty is

$$\begin{aligned} \mathcal{P}_{\mathrm{ACGL}}(E) &= \sum_{j=1}^{d} w_j \|H_K E_{\cdot j}\|_2 = \sum_{j=1}^{d} w_j \|c_{\cdot j}\|_2, \\ w_j^{\mathrm{raw}} &= \left(\|H_K E_{\cdot j}^{\mathrm{D}}\|_2 + \epsilon\right)^{-\gamma}, \qquad w_j = \frac{w_j^{\mathrm{raw}}}{\mathrm{median}_{1\le h\le d}\, w_h^{\mathrm{raw}}}. \end{aligned} \tag{S1}$$

The weights are computed once from the common dense initializer and then held fixed. For $V = E^{(u)} - s\nabla f_t(E^{(u)})$, the weighted proximal step is

$$E_{\cdot j}^{(u+1)} = J_K V_{\cdot j} + \left(1 - \frac{s\lambda_\eta w_j}{\|H_K V_{\cdot j}\|_2}\right)_+ H_K V_{\cdot j}. \tag{S2}$$

The E-step and guard logic are unchanged in form, but the criteria evaluated by the guards are method-specific:

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$$Q^{(\mathrm{ACGL})}_{\lambda_\eta} = Q - \lambda_\eta \sum_{j=1}^{d} w_j \|H_K E_{\cdot j}\|_2, \qquad \mathcal{L}^{(\mathrm{ACGL})}_{\lambda_\eta} = \ell - \lambda_\eta \sum_{j=1}^{d} w_j \|H_K E_{\cdot j}\|_2.$$

The reported analyses use $\gamma = 1$ and $\epsilon = 10^{-6}$; the weights are not updated along the path. This construction applies the adaptive lasso and adaptive group-lasso principles of Zou (2006) and Wang and Leng (2008), respectively, to centered coordinate groups. No oracle-property claim is made.

The final method-specific Karush–Kuhn–Tucker (KKT)-motivated path scale and single path are

$$\begin{aligned}
\widetilde{\lambda}^{\mathrm{ACGL}}_{\max,\eta} &= 1.05 \max_j \frac{\|(H_K G_0)_{\cdot j}\|_2}{w_j}, \\
\Lambda_{\mathrm{final}} &= \{0\} \cup \mathrm{Geom}\left(10^{-4}\widetilde{\lambda}^{\mathrm{ACGL}}_{\max,\eta}, \widetilde{\lambda}^{\mathrm{ACGL}}_{\max,\eta}; 239\right), \\
\Lambda^{\mathrm{hist}}_{\mathrm{std}} &= \{0\} \cup \mathrm{Geom}\left(10^{-2}\widetilde{\lambda}^{\mathrm{ACGL}}_{\max,\eta}, \widetilde{\lambda}^{\mathrm{ACGL}}_{\max,\eta}; 239\right).
\end{aligned} \tag{S3}$$

Here E_0^{com} is the matrix that repeats the pooled common natural parameter, τ^0 contains the responsibilities from the guarded zero-penalty fit, and $G_0 = \nabla f_{\tau^0}(E_0^{\mathrm{com}})$, as defined for the E-CGL path scale in the main article. The final and historical paths start from the same guarded zero-penalty fit obtained from the common dense pilot. The final path processes its positive tuning values in increasing order and uses the same support-deduplication, target-preserving refit, explicit empty-support candidate, and BIC rule as E-CGL. When several valid path fits generate the same support, the fit with the largest observed log-likelihood initializes that support's refit.

Here $\mathrm{Geom}(a, b; q)$ denotes q positive log-geometric values from a through b. The 10^{-2} path and the earlier two-segment audit that adjoined values down to 10^{-6} are retained below as numerical provenance. They motivated replacing the earlier lower endpoint by the single 10^{-4} path, but they do not define the final estimator.

S2. Simulation design and complete numerical results

The primary simulation contains 12 cells formed by $n \in \{300, 1000\}$, target oracle Bayes error $e_B \in \{0.025, 0.05, 0.10\}$, and equal or heterogeneous concentration. Each cell has 100 paired repetitions and six common methods, giving 7,200 method-repetition rows. The recorded output contains no error rows and no duplicate cell-repetition-method keys. Among the 7,200 primary-grid rows, one dense shared-concentration fit in the equal-concentration $e_B = 0.10$, $n = 300$ cell reached its iteration limit. In the additional pure-concentration diagnostic, four dense free-concentration pilots reached their iteration limits. All selected E-series fits passed the recorded majorization, line-search acceptance, and path-endpoint checks. The largest oracle-error calibration discrepancy was 0.172 percentage points (0.00172 on the probability scale). The centered mean-direction group lasso (M-CGL) was fitted as a directional companion in the four prespecified $e_B = 0.05$ cells, adding 400 valid method-repetition rows. It was not subsequently fitted to the remaining eight primary cells.

S2.1. Oracle-error-calibrated data generation and randomization

Let $q_{\text{block}} = 4$ denote the number of decision coordinates assigned to each of the $K = 4$ component blocks. For a decision coordinate in block g, define

$$u_{kj} = \begin{cases} 1, & k = g, \\ -1/(K-1), & k \neq g. \end{cases}$$

For a candidate common-background norm A, the natural parameters are generated as

$$\eta_{kj}(A) = \begin{cases} A/\sqrt{q_C}, & j \in S_C, \\ \alpha_k(A) u_{kj}, & j \in S_D, \\ 0, & j \in S_N, \end{cases} \qquad \alpha_k(A) = \frac{\sqrt{\kappa_k^2 - A^2}}{\|u_{k\cdot}\|_2}.$$

Thus $\|\eta_k(A)\|_2 = \kappa_k$, and $\mu_k = \eta_k/\kappa_k$. Mixture proportions are $1/K$.

The value of A is chosen by 18 bisection steps on $[0, 0.995 \min_k \kappa_k]$. A common-random-number Monte Carlo sample of size 50,000 is used during bisection, followed by an independent validation sample of size 200,000. The final achieved errors are reported in the main article. For target e_B and scenario index $h \in \{1, 2\}$ (equal, heterogeneous), the historical DGP calibration uses

$$s_{\text{cal}} = 271828182 + 100 \text{ round}(1000 e_B) + h.$$

For the final rerun, let $c = 1, \ldots, 24$ denote the fixed row index in the archived cell manifest and let $b = 1, \ldots, 100$ denote the repetition. The production seeds are

$$\begin{aligned} s_{\text{data},c,b} &= 260731 + 100000c + b, \\ s_{\text{test},c,b} &= 260731 + 100000c + 50000 + b, \\ s_{\text{init},c,b} &= 260731 + 100000c + 80000 + b. \end{aligned}$$

These three seed ranges are disjoint across roles, cells, and repetitions. The calibrated data-generating process (DGP) parameters are held fixed across $n = 300$ and $n = 1000$ within each target-error and concentration combination. Within a repetition, all methods are evaluated on the same sample and use the same initialization seed. The likelihood-based mixture methods use the prespecified 10-start initialization protocol appropriate to their concentration parameterization.

Monte Carlo standard errors (MCSEs) were computed from the 100 replicates in each cell. Across all methods and primary cells, their maxima were 0.0094 for $F_{1,\eta}$, 0.0114 for ARI, and 0.101 for MSE_{η^c}. Restricted to E-CGL and E-ACGL, the corresponding maxima were 0.0094, 0.0053, and 0.0375.

The primary selector is BIC after target-specific support refit. For E-CGL and E-ACGL, every distinct path support and the explicit empty support are refitted under an equality constraint that preserves the centered natural-parameter target. The E-series nominal dimension is

$$\mathrm{df}_E(S) = d + (K-1)m + (K-1)\mathbf{1}(m > 0), \qquad m = |S|. \tag{S4}$$

It is a practical selection rule rather than an exact effective degrees of freedom for a nonregular penalized mixture.

The Rossi-style entrywise mean-direction lasso (M-L) proposed by Rossi and Barbaro (2022) uses its event-driven path with at most 240 path points, including the zero-penalty fit, in the primary simulations. E-CGL and E-ACGL use the zero-augmented 240-point KKT-motivated path in the main article. The M-CGL companion uses the separate 240-point dense-score-proxy path defined in Section S11. The E-series selector compares the distinct refitted path supports with the all-common model $S_\eta = \varnothing$. E-CGL and E-ACGL use the target-preserving support-constrained refit and the nominal dimension in (S4). For the prototype-entry method M-L, an M-series mask with $m_k \geq 1$ active entries in component k uses the operational dimension rule

$$\mathrm{df}_{M,\mathrm{free}} = (K-1) + K + \sum_{k=1}^{K} \max\{1, m_k - 1\},$$

for free component-specific concentrations, and

$$\mathrm{df}_{M,\mathrm{shared}} = (K-1) + 1 + \sum_{k=1}^{K} \max\{1, m_k - 1\}$$

for a shared concentration. The floor $\max\{1, m_k - 1\}$ is the operational BIC convention retained for this comparator. Thus these expressions are practical method-specific dimensions rather than exact local dimensions; they approximately account for mixing proportions, concentration parameters, and unit-norm direction constraints. They are not the centered M-CGL dimension, which is defined in Section S11. Dense free- and shared-concentration models have $Kd + K - 1$ and Kd degrees of freedom, respectively. E-ACGL weights are fixed from the common dense component-specific-concentration pilot shared by the two E-series methods, with $\gamma = 1$ and $\epsilon = 10^{-6}$; they are not updated along the path.

Although M-L penalizes component entries, the reported S_P metrics collapse those entries to their coordinate union. They therefore assess prototype-union recovery rather than recovery of the full component-by-coordinate entry mask.

The finite vMF mixture likelihood can be unbounded as a component concentration diverges. Accordingly, dense initialization, the E-CGL/E-ACGL path, and their support-constrained refits use the same finite parameter space,

$$0 \le \kappa_k \le \texttt{kappa\_cap} = 10^6.$$

Proximal proposals outside this space trigger step halving and are recomputed, rather than projected. The support-constrained refit begins from a projected and, if necessary, globally scaled source matrix, then step-halves each unconstrained conditional proposal until the cap and likelihood guard hold. A refit reaching at least $0.99\,\texttt{kappa\_cap}$ is retained for diagnosis but is ineligible for information-criterion selection. This bound defines the numerical problem and does not act as a sparsity penalty.

S2.2. E-series numerical controls

For each generalized M-step, the proximal-gradient loop stops when

$$\frac{\|E^{(u+1)} - E^{(u)}\|_F}{\max\{1, \|E^{(u)}\|_F\}} < 10^{-8}$$

or after 200 iterations in the numerical studies and 160 iterations in the real-data analyses. The smooth majorization tolerance is 10^{-10}, and at most 40 backtracking attempts are allowed. The outer penalized generalized expectation-maximization (GEM) loop uses relative objective tolerance 10^{-7} and initially allows 100 iterations. Auxiliary- and observed-objective decreases larger than 10^{-8} mark the current path point as failed. Nonfinite likelihoods, objectives, or natural parameters and concentration-cap violations are treated likewise. A point that reaches its iteration limit is logged as unconverged but remains in the provisional candidate-support family. An isolated failed point is skipped, and the next tuning value is initialized from the last accepted fit. Three consecutive numerical failures terminate the remaining path without declaring convergence. If the penalized source of the provisional refit-BIC winner is unconverged, the full path is rebuilt with 440 additional iterations per point, only converged penalized sources are retained, and support-constrained refitting and BIC selection are repeated. The path also stops after storing a support with at most one active coordinate.

The support-constrained refit first allows 160 outer iterations and, if the fit is finite, numerically valid, and unconverged, allows 440 additional iterations in the recorded simulation and real-data protocols. Each conditional refit M-step uses the limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm with bounds (L-BFGS-B) (Byrd et al., 1995) and is limited to 80 iterations with projected-gradient tolerance 10^{-6}. The refit stops when the absolute log-likelihood increment is at most $10^{-7} + 10^{-8}|\ell_{\text{old}}|$. These values are numerical controls, not statistical tuning parameters. All reported fits use internally generated dense multistarts rather than the optional externally supplied initialization, and every distinct support is refitted; the optional refit shortlist is not used.

Table S1 distinguishes settings used to obtain the reported results from current software defaults. The numerical studies used 10 dense starts and the real-data analyses used 30. A later increase of the package default for the additional refit budget from 440 to 840 iterations is prospective and does not redefine any recorded fit.

Table S1: Provenance of E-series numerical controls. The two recorded columns define the analyses reported in the article; the last column records current software defaults for settings controlled by `etaVmf`.

Control	Numerical studies	Real data	Current package default
Path points (zero included)	240	240	analysis supplied
Relative positive-path floor	10^{-4}	10^{-4}	10^{-4}
Initial outer penalized GEM iterations	100	100	100
Additional selected-source path iterations	440	440	440
Inner proximal iterations	200	160	200
Backtracking attempts	40	40	40
Initial refit iterations	160	160	160
Additional refit iterations	440	440	840
L-BFGS-B iterations per M-step	80	80	80
L-BFGS-B projected-gradient tolerance	10^{-6}	10^{-6}	10^{-6}
Concentration cap	10^{6}	10^{6}	10^{6}

S2.3. Dense-pilot, selected-source, and path-stopping audits

Across the common 24-cell rerun, 2,396 of the 2,400 dataset-replicate dense free-concentration pilots converged under the 200-iteration limit. Each pilot was shared by E-CGL and E-ACGL, so this count is over dataset replicates rather than 4,800 method-level fits. The four remaining pilots occurred in the pure-concentration cell at repetitions 4, 5, 39, and 56. Each terminal iterate was continued under the same tolerance and concentration cap; convergence required 8–60 additional iterations and increased the observed log-likelihood by 0.30–7.72. Repeating the E-CGL and E-ACGL paths and all support-constrained refits from these retry-converged pilots preserved all eight selected supports and their $F_{1,\eta}$ values. The largest absolute BIC change was 0.0264. If these eight sensitivity fits replaced the recorded fits, the pure-concentration-cell mean ARI would change by less than 0.00013 and mean test NLL by less than 0.0012 for either method. The recorded 100-repetition summaries are therefore retained. The final implementation continues a provisional winning pilot that reaches its iteration limit and, if the continuation remains unconverged, falls back to the best converged original start when one exists. Use of a finite unconverged pilot is flagged and occurs only when no converged candidate is available.

The initial candidate archive contained 4,800 E-series method–replicate groups and 328,703 support-constrained refits. Every selected refit was eligible and converged, but this did not establish convergence of the penalized path fit used to generate its support. We therefore reproduced all 4,800 selected penalized sources from the final 10^{-4} paths. Thirty-seven selected sources had reached the original 100-iteration limit, and 32 of their supports

Table S2: Selected-source convergence corrections. "Affected" counts archived winners whose penalized source reached the 100-iteration limit; "support changed" and "q changed" compare the corrected converged-source winner with the archived winner.

Cell	Method	Affected	Support changed	q changed
Pure concentration, $n = 1000$	E-ACGL	1	1	1
Moderately dense, $n = 300$	E-CGL	2	2	2
Moderately dense, $n = 300$	E-ACGL	6	4	4
Moderately dense, $n = 1000$	E-CGL	3	3	3
Moderately dense, $n = 1000$	E-ACGL	1	1	0
Strongly dense, $n = 1000$	E-CGL	18	17	17
Strongly dense, $n = 1000$	E-ACGL	3	3	3
High-dimensional, $n = 300, d = 500$	E-CGL	3	3	2

were not generated by any converged point on the corresponding recorded path.

Only these 37 method–replicate paths were rerun with a 540-iteration allowance, and the candidate family was reconstructed from converged path points only. Every path point in the 37 reruns converged, and every selected support-constrained refit passed the eligibility checks. Thirty-four selected supports changed, including 32 changes in support size. The corrected archive contains the same 8,300 sparse-method rows and 4,800 E-series method–replicate groups, with 328,701 distinct-support refit records. It has no error rows or duplicate keys, and every cell–method summary contains 100 valid replications. The original archive is retained unchanged for provenance; all additional and stress results in the article use the corrected archive.

Table S2 localizes the 37 corrections. None occurred in the primary 12-cell factorial grid. The pure-concentration change affects one E-ACGL replication; the remaining changes occur in stress cells.

The terminal finite path support has size zero or one in every simulation fit. The support-size early stop was audited separately by reproducing 12 representative E-series paths under the initial 100-iteration allowance. The audit includes both methods in the original three conditions, the difficult heterogeneous $e_B = 0.10$, $n = 300$ cell, the $d = 500 > n$ cell, and the final-path E-ACGL replicate with the smallest first-positive retention. Each fit was continued from the first stored support of size at most one through all 240 tuning values. Table S3 reports the results. All 175 remaining points were accepted, no support with more than one coordinate re-entered, and no new support appeared. Fourteen reproduced path points reached their iteration

Table S3: Targeted full-path audit after the support-size early stop. The remaining points are the configured tuning values evaluated after the recorded path had stopped. "Unconv.-only" counts supports generated by an iteration-limit path point and by no converged point on the reproduced path; none is the selected support.

Cell	Method	Stored	Remaining	Unconv.-only	New	$q > 1$ re-entry
Heterogeneous, $e_B = .05$, $n = 1000$	E-CGL	227	13	0	0	0
Heterogeneous, $e_B = .05$, $n = 1000$	E-ACGL	226	14	0	0	0
Pure concentration, $n = 1000$	E-CGL	236	4	4	0	0
Pure concentration, $n = 1000$	E-ACGL	234	6	0	0	0
Moderately dense, $n = 300$	E-CGL	221	19	0	0	0
Moderately dense, $n = 300$	E-ACGL	214	26	0	0	0
Heterogeneous, $e_B = .10$, $n = 300$	E-CGL	224	16	3	0	0
Heterogeneous, $e_B = .10$, $n = 300$	E-ACGL	222	18	1	0	0
High-dimensional, $n = 300, d = 500$	E-CGL	224	16	2	0	0
High-dimensional, $n = 300, d = 500$	E-ACGL	219	21	1	0	0
Crossed support, $n = 400$, rep. 16	E-CGL	229	11	0	0	0
Crossed support, $n = 400$, rep. 16	E-ACGL	229	11	0	0	0

limit and generated 11 supports not reproduced by a converged path point, but none was the ultimately selected support. Reading the continued fits at thresholds 10^{-7}, 10^{-8}, and 10^{-9} gave identical supports. These checks are targeted diagnostics rather than a proof that support paths are nested; they are distinct from the exhaustive 4,800-winner source audit above.

Each recorded Classic3 full-data and split-specific E-series path contains all 240 configured points and ends with support size 2–4; early stopping therefore truncates none. Of their 2,880 path points, 66 reached the original 100-iteration limit, but none generated a selected refit's source support. Removing them leaves every selected Classic3 support unchanged. All selected Classic3 refits are eligible and converged, and all path points pass the recorded majorization check.

S2.4. Estimand-separation and stress data-generating models

The pure-concentration model has $d = 200$ and a common unit direction supported on the first 16 coordinates with alternating signs. Its concentrations are

$$\kappa = (10, 30, 80, 200),$$

so $|S_\mu| = 0$ and $|S_\eta| = 16$. The shared-natural-parameter model uses the construction above with

$$(q_C, q_D, q_N) = (80, 20, 100), \qquad \kappa = (30, 40, 50, 60),$$

and calibrated common norm $A = 21.0118$ for target $e_B = 0.05$.

The crossed-support diagnostic has $d = 24$ and $\kappa = (30, 40, 50, 60)$. Four η-only coordinates use a common directional loading of 0.20, four μ-only coordinates use a common natural-parameter loading of 8, eight coordinates complete the component row norms with component-specific signed contrasts, and eight coordinates are null. Thus $|S_\mu| = |S_\eta| = 12$, with four coordinates unique to each target and eight in their intersection.

The moderately and strongly dense controls use $d = 200$, common norm $A = 27.7349$, and

$$\kappa = (30, 40, 50, 60),$$

with

$$(q_C, q_D, q_N) = (4, 80, 116) \quad \text{or} \quad (4, 160, 36),$$

respectively. Both were calibrated to an oracle Bayes error of approximately 0.10. The high-dimensional cell uses $n = 300$, $d = 500$, $(q_C, q_D, q_N) = (10, 40, 450)$, $\kappa = (45, 60, 75, 90)$, and calibrated $A = 28.5773$ for target $e_B = 0.05$. The beta-min cell uses $n = 1000$, $d = 200$, $(q_C, q_D, q_N) = (4, 16, 180)$, $\kappa = (30, 40, 50, 60)$, and calibrated $A = 21.1746$. Within each four-coordinate decision block, one contrast has unit weight and three have weight 0.25. These constructions and their invariant checks are the recorded models used by the reported simulations.

S2.5. Partition, support, and parameter metrics

For the true partition $\mathcal{U}$ and fitted partition $\mathcal{V}$, the reported NMI uses geometric-mean entropy normalization,

$$\mathrm{NMI}(\mathcal{U}, \mathcal{V}) = \frac{I(\mathcal{U}; \mathcal{V})}{\{H(\mathcal{U})H(\mathcal{V})\}^{1/2}}.$$

It is set to one when the two partitions are identical and both empirical entropies vanish, and to zero when exactly one empirical entropy vanishes. ARI is computed in its standard chance-adjusted form.

For an estimated coordinate support $\widehat{S}$, the reported decomposition is $\widehat{q}_C = |\widehat{S} \cap S_C|$, $\widehat{q}_D = |\widehat{S} \cap S_D|$, and $\widehat{q}_N = |\widehat{S} \cap S_N|$. The posterior-score contrast support metrics are

$$\mathrm{TPR}_\eta = \frac{\widehat{q}_D}{q_D}, \qquad \mathrm{FPR}_\eta = \frac{\widehat{q}_C + \widehat{q}_N}{q_C + q_N}, \qquad \mathrm{Precision}_\eta(\widehat{S}) = \frac{\widehat{q}_D}{|\widehat{S}|},$$

with precision set to zero for an empty support. When the denominator is positive,

$$F_{1,\eta} = \frac{2\,\mathrm{TPR}_\eta\ \mathrm{Precision}_\eta(\widehat{S})}{\mathrm{TPR}_\eta + \mathrm{Precision}_\eta(\widehat{S})}.$$

When the true support is nonempty and the estimated support is empty, TPR, precision, and $F_{1,\eta}$ are set to zero. More generally, $F_{1,\eta} = 0$ whenever $\mathrm{TPR}_\eta + \mathrm{Precision}_\eta(\widehat{S}) = 0$. For a nonempty true support, exact recovery is recorded by $\mathrm{Exact}_\eta(\widehat{S}) = \mathbf{1}\{\widehat{S} = S^\star_\eta\}$, and the exact-support rate is its Monte Carlo mean. When the true support is empty, these three quantities are undefined and displayed as dashes; exact-empty recovery is reported instead. Target-specific prototype and directional evaluations replace the subscript η by P and μ, respectively. Before parameter MSE is computed, all $K!$ component permutations are examined and the permutation minimizing centered natural-parameter MSE is used for μ, κ, and centered-η comparisons.

Using $(\eta^c)_{kj} = c_{kj}$ for a centered natural-parameter entry, write $c^\star_{kj} = \eta^\star_{kj} - K^{-1}\sum_h \eta^\star_{hj}$ and $\widehat{c}_{kj} = \widehat{\eta}_{kj} - K^{-1}\sum_h \widehat{\eta}_{hj}$. The parameter error reported throughout the article and Supplementary Material is

$$\mathrm{MSE}_{\eta^c} = \frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}(\widehat{c}_{kj} - c^\star_{kj})^2.$$

It is an error for centered natural-parameter contrasts, rather than for the uncentered natural parameters.

Tables S4–S9 provide the complete primary-grid, sample-size, oracle-benchmark, stress, and numerical diagnostic results summarized in the main article. The machine-readable companion file `supplementary_simulation_summary.csv` contains the method–cell means and standard deviations used for the numerical tables, including the final single-path E-ACGL summaries for the 12 primary and five stress cells and the target-specific support metrics needed to reproduce Monte Carlo standard errors.

S3. Full primary-factorial results

The following tables report all primary-factorial method–cell combinations. M-CGL was prespecified only for the four $e_B = 0.05$ companion cells. Methods without a coordinate selector have no support entries. In equal-κ cells, the recorded canonical output aggregates the four common coordinates and 180 zero-noise coordinates in one null category; their separate selected counts are therefore not reconstructed.

Table S4: Full primary-factorial support results. Values are means over 100 paired repetitions.

n	e_B	κ	method	$\widehat{q}$	$\widehat{q}_C$	$\widehat{q}_D$	$\widehat{q}_N$	TPR_η	FPR_η	$F_{1,\eta}$	exact
300	0.025	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.025	equal	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.025	equal	Spherical k-means	–	–	–	–	–	–	–	–
300	0.025	equal	E-ACGL	16.14	–	16.00	–	1.000	0.001	0.996	0.860
300	0.025	equal	E-CGL	16.14	–	16.00	–	1.000	0.001	0.996	0.860
300	0.025	equal	M-L	199.58	–	16.00	–	1.000	0.998	0.148	0.000
300	0.025	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.025	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.025	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
300	0.025	hetero.	E-ACGL	16.15	0.00	16.00	0.15	1.000	0.001	0.996	0.880
300	0.025	hetero.	E-CGL	16.15	0.00	16.00	0.15	1.000	0.001	0.996	0.880
300	0.025	hetero.	M-L	199.61	4.00	16.00	179.61	1.000	0.998	0.148	0.000
300	0.050	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.050	equal	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.050	equal	Spherical k-means	–	–	–	–	–	–	–	–
300	0.050	equal	E-ACGL	16.23	–	16.00	–	1.000	0.001	0.993	0.790
300	0.050	equal	E-CGL	16.25	–	16.00	–	1.000	0.001	0.992	0.780
300	0.050	equal	M-CGL	16.24	–	16.00	–	1.000	0.001	0.993	0.780
300	0.050	equal	M-L	199.65	–	16.00	–	1.000	0.998	0.148	0.000
300	0.050	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.050	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.050	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
300	0.050	hetero.	E-ACGL	16.18	0.00	16.00	0.18	1.000	0.001	0.995	0.850
300	0.050	hetero.	E-CGL	16.67	0.01	16.00	0.66	1.000	0.004	0.981	0.670
300	0.050	hetero.	M-CGL	20.90	3.65	14.68	2.57	0.917	0.034	0.797	0.000
300	0.050	hetero.	M-L	199.70	4.00	16.00	179.70	1.000	0.998	0.148	0.000
300	0.100	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.100	equal	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.100	equal	Spherical k-means	–	–	–	–	–	–	–	–
300	0.100	equal	E-ACGL	16.33	–	15.95	–	0.997	0.002	0.987	0.770
300	0.100	equal	E-CGL	16.28	–	16.00	–	1.000	0.002	0.992	0.770
300	0.100	equal	M-L	199.67	–	16.00	–	1.000	0.998	0.148	0.000
300	0.100	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
300	0.100	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
300	0.100	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
300	0.100	hetero.	E-ACGL	15.23	0.00	14.80	0.43	0.925	0.002	0.947	0.220
300	0.100	hetero.	E-CGL	20.97	0.13	14.81	6.03	0.926	0.033	0.816	0.020
300	0.100	hetero.	M-L	199.62	4.00	16.00	179.62	1.000	0.998	0.148	0.000
1000	0.025	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.025	equal	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.025	equal	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.025	equal	E-ACGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
1000	0.025	equal	E-CGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
1000	0.025	equal	M-L	199.42	–	16.00	–	1.000	0.997	0.149	0.000
1000	0.025	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.025	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.025	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.025	hetero.	E-ACGL	16.02	0.01	16.00	0.01	1.000	0.000	0.999	0.980
1000	0.025	hetero.	E-CGL	16.02	0.01	16.00	0.01	1.000	0.000	0.999	0.980
1000	0.025	hetero.	M-L	199.57	4.00	16.00	179.57	1.000	0.998	0.148	0.000
1000	0.050	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.050	equal	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.050	equal	Spherical k-means	–	–	–	–	–	–	–	–

n	e_B	κ	method	$\widehat{q}$	$\widehat{q}_C$	$\widehat{q}_D$	$\widehat{q}_N$	TPR$_\eta$	FPR$_\eta$	$F_{1,\eta}$	exact
1000	0.050	equal	E-ACGL	16.06	–	16.00	–	1.000	0.000	0.998	0.950
1000	0.050	equal	E-CGL	16.06	–	16.00	–	1.000	0.000	0.998	0.950
1000	0.050	equal	M-CGL	16.04	–	16.00	–	1.000	0.000	0.999	0.960
1000	0.050	equal	M-L	199.58	–	16.00	–	1.000	0.998	0.148	0.000
1000	0.050	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.050	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.050	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.050	hetero.	E-ACGL	16.02	0.00	16.00	0.02	1.000	0.000	0.999	0.980
1000	0.050	hetero.	E-CGL	16.02	0.00	16.00	0.02	1.000	0.000	0.999	0.980
1000	0.050	hetero.	M-CGL	20.04	4.00	16.00	0.04	1.000	0.022	0.888	0.000
1000	0.050	hetero.	M-L	199.67	4.00	16.00	179.67	1.000	0.998	0.148	0.000
1000	0.100	equal	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.100	equal	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.100	equal	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.100	equal	E-ACGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
1000	0.100	equal	E-CGL	16.02	–	16.00	–	1.000	0.000	0.999	0.980
1000	0.100	equal	M-L	199.64	–	16.00	–	1.000	0.998	0.148	0.000
1000	0.100	hetero.	Dense free-κ_k	–	–	–	–	–	–	–	–
1000	0.100	hetero.	Dense shared-κ	–	–	–	–	–	–	–	–
1000	0.100	hetero.	Spherical k-means	–	–	–	–	–	–	–	–
1000	0.100	hetero.	E-ACGL	16.01	0.00	16.00	0.01	1.000	0.000	1.000	0.990
1000	0.100	hetero.	E-CGL	16.41	0.01	16.00	0.40	1.000	0.002	0.988	0.740
1000	0.100	hetero.	M-L	199.88	4.00	16.00	179.88	1.000	0.999	0.148	0.000

Table S5: Full primary-factorial clustering, prediction, and estimation results. Values are means; MCSEs are retained in the accompanying CSV.

n	e_B	κ	method	ARI	NMI	test NLL	MSE$_{\eta^c}$	valid	fail rate
300	0.025	equal	Dense free-κ_k	0.864	0.812	-245.906	2.505	100	0.000
300	0.025	equal	Dense shared-κ	0.869	0.816	-245.921	2.471	100	0.000
300	0.025	equal	Spherical k-means	0.863	0.810	–	–	100	0.000
300	0.025	equal	E-ACGL	0.927	0.888	-246.935	0.198	100	0.000
300	0.025	equal	E-CGL	0.927	0.888	-246.935	0.198	100	0.000
300	0.025	equal	M-L	0.865	0.814	-245.934	2.447	100	0.000
300	0.025	hetero.	Dense free-κ_k	0.856	0.820	-246.110	2.540	100	0.000
300	0.025	hetero.	Dense shared-κ	0.798	0.767	-245.869	3.092	100	0.000
300	0.025	hetero.	Spherical k-means	0.775	0.746	–	–	100	0.000
300	0.025	hetero.	E-ACGL	0.925	0.894	-247.153	0.192	100	0.000
300	0.025	hetero.	E-CGL	0.925	0.894	-247.153	0.192	100	0.000
300	0.025	hetero.	M-L	0.860	0.824	-246.138	2.475	100	0.000
300	0.050	equal	Dense free-κ_k	0.726	0.669	-245.919	2.863	100	0.000
300	0.050	equal	Dense shared-κ	0.738	0.676	-245.936	2.810	100	0.000
300	0.050	equal	Spherical k-means	0.620	0.569	–	–	100	0.000
300	0.050	equal	E-ACGL	0.856	0.802	-247.011	0.219	100	0.000
300	0.050	equal	E-CGL	0.856	0.802	-247.010	0.220	100	0.000
300	0.050	equal	M-CGL	0.856	0.802	-247.009	0.222	100	0.000
300	0.050	equal	M-L	0.731	0.674	-245.954	2.771	100	0.000
300	0.050	hetero.	Dense free-κ_k	0.709	0.690	-246.082	3.077	100	0.000
300	0.050	hetero.	Dense shared-κ	0.649	0.631	-245.860	3.764	100	0.000
300	0.050	hetero.	Spherical k-means	0.580	0.570	–	–	100	0.000
300	0.050	hetero.	E-ACGL	0.858	0.822	-247.215	0.209	100	0.000
300	0.050	hetero.	E-CGL	0.857	0.821	-247.205	0.232	100	0.000
300	0.050	hetero.	M-CGL	0.815	0.789	-247.064	0.615	100	0.000

n	e_B	κ	method	ARI	NMI	test NLL	MSE$_{\eta^c}$	valid	fail rate
300	0.050	hetero.	M-L	0.718	0.698	-246.122	2.957	100	0.000
300	0.100	equal	Dense free-κ_k	0.390	0.371	-245.737	4.509	100	0.000
300	0.100	equal	Dense shared-κ	0.436	0.397	-245.788	4.191	100	0.000
300	0.100	equal	Spherical k-means	0.211	0.202	–	–	100	0.000
300	0.100	equal	E-ACGL	0.721	0.661	-247.124	0.273	100	0.000
300	0.100	equal	E-CGL	0.724	0.664	-247.130	0.253	100	0.000
300	0.100	equal	M-L	0.422	0.400	-245.830	4.143	100	0.000
300	0.100	hetero.	Dense free-κ_k	0.512	0.530	-246.148	4.591	100	0.000
300	0.100	hetero.	Dense shared-κ	0.484	0.481	-245.998	4.768	100	0.000
300	0.100	hetero.	Spherical k-means	0.444	0.441	–	–	100	0.000
300	0.100	hetero.	E-ACGL	0.702	0.686	-247.283	0.419	100	0.000
300	0.100	hetero.	E-CGL	0.636	0.666	-247.164	0.871	100	0.000
300	0.100	hetero.	M-L	0.514	0.535	-246.173	4.529	100	0.000
1000	0.025	equal	Dense free-κ_k	0.917	0.875	-246.964	0.679	100	0.000
1000	0.025	equal	Dense shared-κ	0.918	0.875	-246.966	0.676	100	0.000
1000	0.025	equal	Spherical k-means	0.917	0.874	–	–	100	0.000
1000	0.025	equal	E-ACGL	0.933	0.895	-247.245	0.055	100	0.000
1000	0.025	equal	E-CGL	0.933	0.895	-247.245	0.055	100	0.000
1000	0.025	equal	M-L	0.918	0.875	-246.968	0.672	100	0.000
1000	0.025	hetero.	Dense free-κ_k	0.912	0.879	-247.167	0.685	100	0.000
1000	0.025	hetero.	Dense shared-κ	0.861	0.830	-246.931	1.147	100	0.000
1000	0.025	hetero.	Spherical k-means	0.866	0.833	–	–	100	0.000
1000	0.025	hetero.	E-ACGL	0.928	0.897	-247.451	0.056	100	0.000
1000	0.025	hetero.	E-CGL	0.928	0.897	-247.451	0.056	100	0.000
1000	0.025	hetero.	M-L	0.913	0.879	-247.170	0.677	100	0.000
1000	0.050	equal	Dense free-κ_k	0.836	0.779	-247.032	0.737	100	0.000
1000	0.050	equal	Dense shared-κ	0.837	0.780	-247.035	0.731	100	0.000
1000	0.050	equal	Spherical k-means	0.835	0.777	–	–	100	0.000
1000	0.050	equal	E-ACGL	0.867	0.814	-247.316	0.061	100	0.000
1000	0.050	equal	E-CGL	0.867	0.814	-247.316	0.061	100	0.000
1000	0.050	equal	M-CGL	0.867	0.814	-247.316	0.061	100	0.000
1000	0.050	equal	M-L	0.837	0.779	-247.036	0.726	100	0.000
1000	0.050	hetero.	Dense free-κ_k	0.835	0.798	-247.228	0.747	100	0.000
1000	0.050	hetero.	Dense shared-κ	0.765	0.737	-247.004	1.273	100	0.000
1000	0.050	hetero.	Spherical k-means	0.773	0.741	–	–	100	0.000
1000	0.050	hetero.	E-ACGL	0.869	0.831	-247.517	0.061	100	0.000
1000	0.050	hetero.	E-CGL	0.869	0.831	-247.517	0.061	100	0.000
1000	0.050	hetero.	M-CGL	0.867	0.830	-247.510	0.087	100	0.000
1000	0.050	hetero.	M-L	0.836	0.799	-247.232	0.736	100	0.000
1000	0.100	equal	Dense free-κ_k	0.682	0.619	-247.144	0.866	100	0.000
1000	0.100	equal	Dense shared-κ	0.684	0.621	-247.146	0.858	100	0.000
1000	0.100	equal	Spherical k-means	0.676	0.613	–	–	100	0.000
1000	0.100	equal	E-ACGL	0.747	0.684	-247.445	0.067	100	0.000
1000	0.100	equal	E-CGL	0.747	0.684	-247.445	0.067	100	0.000
1000	0.100	equal	M-L	0.683	0.621	-247.149	0.851	100	0.000
1000	0.100	hetero.	Dense free-κ_k	0.669	0.654	-247.308	1.013	100	0.000
1000	0.100	hetero.	Dense shared-κ	0.595	0.586	-247.110	1.758	100	0.000
1000	0.100	hetero.	Spherical k-means	0.589	0.577	–	–	100	0.000
1000	0.100	hetero.	E-ACGL	0.753	0.722	-247.630	0.071	100	0.000
1000	0.100	hetero.	E-CGL	0.753	0.722	-247.627	0.077	100	0.000
1000	0.100	hetero.	M-L	0.675	0.658	-247.316	0.970	100	0.000

S4. Sample-size and oracle-support diagnostics

Table S6 reports the 16 E-CGL sample-size trajectories. For MSE, TPR, and FPR, parentheses contain 100× Monte Carlo standard errors. An oracle-support comparison denotes a fixed-known-support refit; no asymptotic selection claim is made.

S5. Stress conditions

The five stress cells cover moderate density at two sample sizes, a strongly dense negative control, $d > n$, and a weak beta-min condition. Logged method times in the machine-readable table retain their original timing scopes and are not used as matched end-to-end comparisons.

Table S8: Full stress-condition support and fit results.

scenario	n	d	q^*	method	$\widehat{q}$	TPR	FPR	$F_{1,\eta}$	exact	ARI	NLL	MSE_{η^c}
moderate	300	200	80	Dense free-κ_k	–	–	–	–	–	0.512	-246.125	4.393
moderate	300	200	80	Dense shared-κ	–	–	–	–	–	0.481	-245.977	4.830
moderate	300	200	80	Spherical k-means	–	–	–	–	–	0.449	–	–
moderate	300	200	80	E-ACGL	58.15	0.689	0.025	0.797	0.000	0.545	-246.709	3.364
moderate	300	200	80	E-CGL	81.99	0.814	0.141	0.805	0.000	0.556	-246.738	4.985
moderate	300	200	80	M-L	199.50	1.000	0.996	0.572	0.000	0.515	-246.144	4.360
moderate	1000	200	80	Dense free-κ_k	–	–	–	–	–	0.670	-247.313	0.989
moderate	1000	200	80	Dense shared-κ	–	–	–	–	–	0.600	-247.120	1.719
moderate	1000	200	80	Spherical k-means	–	–	–	–	–	0.586	–	–
moderate	1000	200	80	E-ACGL	67.53	0.843	0.001	0.913	0.000	0.679	-247.444	0.642
moderate	1000	200	80	E-CGL	80.31	0.902	0.068	0.901	0.000	0.649	-247.421	0.820
moderate	1000	200	80	M-L	199.81	1.000	0.998	0.572	0.000	0.673	-247.320	0.957
strong dense	1000	200	160	Dense free-κ_k	–	–	–	–	–	0.669	-247.312	0.998
strong dense	1000	200	160	Dense shared-κ	–	–	–	–	–	0.598	-247.120	1.716
strong dense	1000	200	160	Spherical k-means	–	–	–	–	–	0.589	–	–
strong dense	1000	200	160	E-ACGL	125.60	0.783	0.007	0.877	0.000	0.613	-247.276	1.611
strong dense	1000	200	160	E-CGL	152.49	0.902	0.205	0.923	0.000	0.632	-247.307	2.005
strong dense	1000	200	160	M-L	199.73	1.000	0.993	0.890	0.000	0.671	-247.318	0.968
$d > n$	300	500	40	Dense free-κ_k	–	–	–	–	–	0.424	-840.924	9.519
$d > n$	300	500	40	Dense shared-κ	–	–	–	–	–	0.405	-840.769	9.863
$d > n$	300	500	40	Spherical k-means	–	–	–	–	–	0.285	–	–
$d > n$	300	500	40	E-ACGL	36.74	0.853	0.006	0.888	0.000	0.766	-843.754	1.252
$d > n$	300	500	40	E-CGL	54.70	0.927	0.038	0.803	0.000	0.759	-843.626	1.548
$d > n$	300	500	40	M-L	499.99	1.000	1.000	0.148	0.000	0.431	-840.945	9.441
weak beta-min	1000	200	16	Dense free-κ_k	–	–	–	–	–	0.837	-247.224	0.750
weak beta-min	1000	200	16	Dense shared-κ	–	–	–	–	–	0.768	-247.006	1.269
weak beta-min	1000	200	16	Spherical k-means	–	–	–	–	–	0.776	–	–
weak beta-min	1000	200	16	E-ACGL	15.99	0.998	0.000	0.998	0.930	0.869	-247.516	0.059
weak beta-min	1000	200	16	E-CGL	15.99	0.998	0.000	0.998	0.930	0.869	-247.516	0.059
weak beta-min	1000	200	16	M-L	199.72	1.000	0.998	0.148	0.000	0.838	-247.228	0.740

Table S6: E-CGL sample-size trajectories.

e_B	κ	n	MSE_{η^c} (100 × MCSE)	$n\,\mathrm{MSE}_{\eta^c}$	TPR (100 × MCSE)	FPR (100 × MCSE)	$F_{1,\eta}$	$\widehat{q}$	exact	ARI	test NLL
0.050	equal	300	0.220 (0.599)	66.136	1.000 (0.000)	0.001 (0.028)	0.992	16.25	0.780	0.856	-247.010
0.050	equal	600	0.097 (0.215)	58.240	1.000 (0.000)	0.000 (0.008)	0.999	16.02	0.980	0.864	-247.236
0.050	equal	1000	0.061 (0.161)	61.259	1.000 (0.000)	0.000 (0.015)	0.998	16.06	0.950	0.867	-247.316
0.050	equal	2000	0.028 (0.055)	56.450	1.000 (0.000)	0.000 (0.000)	1.000	16.00	1.000	0.869	-247.378
0.050	hetero.	300	0.232 (0.822)	69.470	1.000 (0.000)	0.004 (0.072)	0.981	16.67	0.670	0.857	-247.205
0.050	hetero.	600	0.099 (0.245)	59.680	1.000 (0.000)	0.000 (0.015)	0.998	16.06	0.950	0.866	-247.434
0.050	hetero.	1000	0.061 (0.132)	61.403	1.000 (0.000)	0.000 (0.008)	0.999	16.02	0.980	0.869	-247.517
0.050	hetero.	2000	0.028 (0.061)	55.642	1.000 (0.000)	0.000 (0.000)	1.000	16.00	1.000	0.871	-247.572
0.100	equal	300	0.253 (0.660)	75.779	1.000 (0.000)	0.002 (0.030)	0.992	16.28	0.770	0.724	-247.130
0.100	equal	600	0.110 (0.243)	66.185	1.000 (0.000)	0.000 (0.009)	0.999	16.03	0.970	0.742	-247.359
0.100	equal	1000	0.067 (0.133)	66.562	1.000 (0.000)	0.000 (0.008)	0.999	16.02	0.980	0.747	-247.445
0.100	equal	2000	0.031 (0.063)	62.763	1.000 (0.000)	0.000 (0.000)	1.000	16.00	1.000	0.749	-247.504
0.100	hetero.	300	0.871 (3.751)	261.231	0.926 (0.773)	0.033 (0.344)	0.816	20.97	0.020	0.636	-247.164
0.100	hetero.	600	0.194 (0.991)	116.421	0.993 (0.250)	0.013 (0.141)	0.934	18.20	0.280	0.739	-247.518
0.100	hetero.	1000	0.077 (0.198)	77.101	1.000 (0.000)	0.002 (0.047)	0.988	16.41	0.740	0.753	-247.627
0.100	hetero.	2000	0.035 (0.079)	69.955	1.000 (0.000)	0.000 (0.009)	0.999	16.03	0.970	0.757	-247.686

Table S7: BIC-after-refit, path-oracle support, and oracle-S_η refit comparisons. Dashes indicate metrics not stored for the support-only path oracle. The selector gap is the path-oracle minus BIC-selected $F_{1,\eta}$; the path gap is one minus the path-oracle $F_{1,\eta}$.

e_B	κ	n	selector	$\widehat{q}$	$F_{1,\eta}$	exact	MSE_{η^c}	test NLL	NLL gap	selector gap	path gap
0.050	equal	300	BIC-after-refit	16.25	0.992	0.780	0.220	-247.010	0.008	0.008	–
0.050	equal	300	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	equal	300	oracle-S_η refit	16.00	1.000	1.000	0.201	-247.018	0.000	–	–
0.050	equal	600	BIC-after-refit	16.02	0.999	0.980	0.097	-247.236	0.000	0.001	–
0.050	equal	600	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	equal	600	oracle-S_η refit	16.00	1.000	1.000	0.096	-247.237	0.000	–	–
0.050	equal	1000	BIC-after-refit	16.06	0.998	0.950	0.061	-247.316	0.001	0.002	–
0.050	equal	1000	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	equal	1000	oracle-S_η refit	16.00	1.000	1.000	0.060	-247.317	0.000	–	–
0.050	equal	2000	BIC-after-refit	16.00	1.000	1.000	0.028	-247.378	-0.000	0.000	–
0.050	equal	2000	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	equal	2000	oracle-S_η refit	16.00	1.000	1.000	0.028	-247.378	0.000	–	–
0.050	hetero.	300	BIC-after-refit	16.67	0.981	0.670	0.232	-247.205	0.015	0.008	–
0.050	hetero.	300	path oracle	16.22	0.989	0.760	–	–	–	–	0.011
0.050	hetero.	300	oracle-S_η refit	16.00	1.000	1.000	0.196	-247.220	0.000	–	–
0.050	hetero.	600	BIC-after-refit	16.06	0.998	0.950	0.099	-247.434	0.001	0.002	–
0.050	hetero.	600	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	hetero.	600	oracle-S_η refit	16.00	1.000	1.000	0.097	-247.435	0.000	–	–
0.050	hetero.	1000	BIC-after-refit	16.02	0.999	0.980	0.061	-247.517	0.000	0.001	–
0.050	hetero.	1000	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	hetero.	1000	oracle-S_η refit	16.00	1.000	1.000	0.061	-247.517	0.000	–	–
0.050	hetero.	2000	BIC-after-refit	16.00	1.000	1.000	0.028	-247.572	0.000	0.000	–
0.050	hetero.	2000	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.050	hetero.	2000	oracle-S_η refit	16.00	1.000	1.000	0.028	-247.572	0.000	–	–
0.100	hetero.	300	BIC-after-refit	20.97	0.816	0.020	0.871	-247.164	0.165	0.048	–
0.100	hetero.	300	path oracle	18.89	0.864	0.060	–	–	–	–	0.136
0.100	hetero.	300	oracle-S_η refit	16.00	1.000	1.000	0.345	-247.329	0.000	–	–
0.100	hetero.	600	BIC-after-refit	18.20	0.934	0.280	0.194	-247.518	0.027	0.050	–
0.100	hetero.	600	path oracle	16.09	0.984	0.630	–	–	–	–	0.016
0.100	hetero.	600	oracle-S_η refit	16.00	1.000	1.000	0.118	-247.545	0.000	–	–
0.100	hetero.	1000	BIC-after-refit	16.41	0.988	0.740	0.077	-247.627	0.003	0.011	–
0.100	hetero.	1000	path oracle	16.02	0.999	0.980	–	–	–	–	0.001
0.100	hetero.	1000	oracle-S_η refit	16.00	1.000	1.000	0.070	-247.630	0.000	–	–
0.100	hetero.	2000	BIC-after-refit	16.03	0.999	0.970	0.035	-247.686	0.000	0.001	–
0.100	hetero.	2000	path oracle	16.00	1.000	1.000	–	–	–	–	0.000
0.100	hetero.	2000	oracle-S_η refit	16.00	1.000	1.000	0.034	-247.686	0.000	–	–

Table S9: Stress-condition convergence and logged method-phase timing. Times retain method-specific scopes and are descriptive only.

scenario	n	d	method	valid	convergence	failure	endpoint	median sec.	IQR sec.
moderate	300	200	Dense free-κ_k	100	1.000	0.000	–	3.31	[2.89, 3.73]
moderate	300	200	Dense shared-κ	100	1.000	0.000	–	1.35	[1.13, 1.51]
moderate	300	200	Spherical k-means	100	1.000	0.000	–	0.07	[0.06, 0.08]
moderate	300	200	E-ACGL	100	1.000	0.000	–	110.86	[104.70, 119.02]
moderate	300	200	E-CGL	100	1.000	0.000	–	89.89	[85.10, 97.75]
moderate	300	200	M-L	100	1.000	0.000	–	17.06	[14.88, 19.22]
moderate	1000	200	Dense free-κ_k	100	1.000	0.000	–	11.32	[9.79, 12.66]
moderate	1000	200	Dense shared-κ	100	1.000	0.000	–	4.64	[4.18, 5.23]
moderate	1000	200	Spherical k-means	100	1.000	0.000	–	0.25	[0.22, 0.27]
moderate	1000	200	E-ACGL	100	1.000	0.000	–	109.47	[104.28, 115.96]
moderate	1000	200	E-CGL	100	1.000	0.000	–	87.46	[83.12, 94.08]
moderate	1000	200	M-L	100	1.000	0.000	–	51.40	[46.10, 59.22]
strong dense	1000	200	Dense free-κ_k	100	1.000	0.000	–	10.98	[9.25, 12.71]
strong dense	1000	200	Dense shared-κ	100	1.000	0.000	–	4.50	[3.88, 5.18]
strong dense	1000	200	Spherical k-means	100	1.000	0.000	–	0.24	[0.22, 0.26]
strong dense	1000	200	E-ACGL	100	1.000	0.000	–	110.16	[101.35, 120.01]
strong dense	1000	200	E-CGL	100	1.000	0.000	–	89.02	[82.56, 99.63]
strong dense	1000	200	M-L	100	1.000	0.000	–	54.88	[47.22, 61.58]
$d > n$	300	500	Dense free-κ_k	100	1.000	0.000	–	5.17	[4.71, 5.56]
$d > n$	300	500	Dense shared-κ	100	1.000	0.000	–	0.94	[0.86, 1.00]
$d > n$	300	500	Spherical k-means	100	1.000	0.000	–	0.09	[0.08, 0.11]
$d > n$	300	500	E-ACGL	100	1.000	0.000	–	219.15	[206.81, 237.74]
$d > n$	300	500	E-CGL	100	1.000	0.000	–	181.00	[172.00, 197.93]
$d > n$	300	500	M-L	100	1.000	0.000	–	31.41	[29.77, 34.23]
weak beta-min	1000	200	Dense free-κ_k	100	1.000	0.000	–	4.66	[3.96, 5.37]
weak beta-min	1000	200	Dense shared-κ	100	1.000	0.000	–	1.56	[1.40, 1.81]
weak beta-min	1000	200	Spherical k-means	100	1.000	0.000	–	0.25	[0.23, 0.28]
weak beta-min	1000	200	E-ACGL	100	1.000	0.000	–	82.50	[79.98, 85.55]
weak beta-min	1000	200	E-CGL	100	1.000	0.000	–	67.85	[65.22, 69.84]
weak beta-min	1000	200	M-L	100	1.000	0.000	–	50.78	[47.31, 53.22]

S5.1. Empty-support path sensitivity

The E-series candidate family includes the all-common model even when the finite path does not generate it. Under the historical 10^{-2}-floor path, we evaluated this candidate retrospectively for both E-series methods in 24 simulation cells, giving 4,800 method–repetition comparisons. It changed 84 selections, all in the three stress combinations in Table S10; primary factorial, estimand-separation, and sample-size results were unchanged. These rows document the diagnostic that motivated the wider candidate path and are not final performance results. On the final single 10^{-4}-floor path, neither E-CGL nor E-ACGL selected the empty support in any of the 2,400 recorded method–repetition fits per method in this common 24-cell rerun subset.

Table S10: Historical 10^{-2}-floor audit of the explicit empty-support candidate. Rows list the stress cells in which at least one selected model changed; these counts are retained as path-development provenance.

Stress condition	Method	Comparisons	Empty-support wins
Moderately dense, $n = 300, d = 200, q_D = 80$	E-CGL	100	81
Moderately dense, $n = 300, d = 200, q_D = 80$	E-ACGL	100	1
High-dimensional, $n = 300, d = 500, q_D = 40$	E-CGL	100	2

S6. Matched penalty-structure ablation

E-CL denotes the entrywise centered lasso with

$$\mathcal{P}_{\mathrm{E-CL}}(E) = \sum_{j=1}^{d} \sum_{k=1}^{K} |c_{kj}|, \qquad c_{\cdot j} = H_K E_{\cdot j}.$$

The fitted E-CL criterion subtracts $\lambda_\eta \mathcal{P}_{\mathrm{E-CL}}(E)$.

E-CL, E-CGL, and E-ACGL were compared using the matched smooth objective, paired samples and dense initializations, method-specific KKT endpoints, 120-point paths, and target-preserving support-constrained refits. Every distinct path support was refitted and compared by df_E-based BIC after refit. Each cell used 100 repetitions (Table S11). The quantity $\widehat{q}_{\mathrm{part}}$ counts coordinates for which the penalized source solution retained only a proper subset of the K centered component entries; it is evaluated before support refitting.

The well-separated S1 cell has $K = 4$, $n = 1000$, $d = 200$, and $q_D = 16$. The difficult cell has $K = 4$, $n = 300$, $d = 200$, $(q_C, q_D, q_N) = (4, 16, 180)$, heterogeneous concentrations, and target $e_B = 0.10$.

Table S11: Matched ablation of centered penalty structures. Entries are means over 100 paired replications per cell.

Cell	Method	$\widehat{q}$	$\widehat{q}_C$	$\widehat{q}_D$	$\widehat{q}_N$	$\widehat{q}_{\text{part}}$	$F_{1,\eta}$	ARI	MSE_{η^c}
S1	E-CL	16.00	0.00	16.00	0.00	1.41	1.000	0.875	0.058
S1	E-CGL	16.00	0.00	16.00	0.00	0.00	1.000	0.875	0.058
S1	E-ACGL	16.00	0.00	16.00	0.00	0.00	1.000	0.875	0.058
difficult $n = 300$ cell	E-CL	26.51	0.22	15.09	11.20	26.23	0.737	0.606	1.132
difficult $n = 300$ cell	E-CGL	20.28	0.10	14.70	5.48	0.00	0.822	0.633	0.923
difficult $n = 300$ cell	E-ACGL	15.32	0.00	14.85	0.47	0.00	0.948	0.706	0.413

The well-separated S1 cell did not distinguish the three penalties by support recovery. In the difficult cell, E-CGL selected 5.72 fewer noise coordinates than E-CL and increased $F_{1,\eta}$ by 0.085; the paired Monte Carlo standard errors were 0.82 and 0.010, respectively. E-ACGL selected 5.01 fewer noise coordinates than E-CGL and increased $F_{1,\eta}$ by 0.126, with paired Monte Carlo standard errors 0.53 and 0.009. Thus, coordinate grouping reduced entry fragmentation and false selection in the difficult cell, while adaptive weighting provided an additional improvement. The ablation isolates the effect of fixed adaptive weighting and does not imply uniform superiority over all regimes.

S7. Selector sensitivity

A proximal-path sensitivity diagnostic compared BIC before and after refitting. It used the main nominal dimension $\text{df}_A(S) := \text{df}_E(S)$ in (S4), and the active-only sensitivity dimension

$$\text{df}_B = Km + (K - 1)\mathbf{1}(m > 0), \qquad m = |S|.$$

For support selection, the EBIC sensitivity criterion (Chen and Chen, 2008) was implemented as

$$\begin{aligned}\mathrm{EBIC}_\gamma(S) = -2\ell\Big(\widehat{\Theta}_S^{\mathrm{refit}}\Big) &+ \log(n)\,\mathrm{df}_A(S)\\ &+ 2\gamma\log\binom{d}{|S|}, \qquad \gamma \in \{0.25, 0.5, 1\}.\end{aligned}$$

The two dimension rules are practical penalty slopes rather than exact effective degrees of freedom. Within this five-replication audit, $\gamma = 0.25$ and 0.5 selected identical supports within each method. For E-CGL, both gave mean $\widehat{q} = 17.2$, $\widehat{q}_D = 13.8$, $\widehat{q}_N = 3.4$, $F_{1,\eta} = 0.834$, and ARI $= 0.631$; for E-ACGL, both selected the same supports as BIC after refitting under df_A. Table S12 shows the most difficult $n = 300$ cell, displays $\gamma = 1$ as the strongest EBIC setting, and omits the duplicate intermediate rows. This five-replication diagnostic is interpreted as a selector audit rather than a performance experiment.

Table S12: Sensitivity to the information criterion in the difficult heterogeneous cell. Entries are means over five diagnostic replications.

Method	Rule	$\widehat{q}$	$\widehat{q}_D$	$\widehat{q}_N$	$F_{1,\eta}$	ARI
E-CGL	BIC after, df-A	19.0	14.2	4.8	0.814	0.655
E-CGL	BIC before, df-A	14.6	13.4	1.2	0.874	0.627
E-CGL	BIC after, df-B	14.8	13.6	1.2	0.882	0.616
E-CGL	EBIC_1 after, df-A	14.8	13.6	1.2	0.882	0.616
E-ACGL	BIC after, df-A	14.8	14.6	0.2	0.945	0.692
E-ACGL	BIC before, df-A	15.4	14.4	1.0	0.916	0.694
E-ACGL	BIC after, df-B	14.2	14.2	0.0	0.937	0.684
E-ACGL	EBIC_1 after, df-A	14.2	14.2	0.0	0.937	0.684

The stronger criteria removed false positives but also removed posterior-score coordinates in the difficult cell. BIC after target-preserving support-constrained refit with df_A is retained as the primary practical rule; no exact effective-df claim is made.

Table S13 separates two events that a single exact-support recovery rate conflates. Path coverage asks whether the fitted path ever contained the true support; conditional BIC success asks whether BIC selected it when it was available. In the difficult heterogeneous trajectory, the principal small-sample limitation was incomplete path coverage. The $e_B = 0.10$ selector gap

Table S13: Separation of path coverage and conditional BIC success for E-CGL. Path coverage is the proportion of repetitions in which the true support occurred among fitted path candidates. Conditional success is computed only among those repetitions.

e_B	κ pattern	n	path coverage	BIC exact	BIC exact | covered
0.05	equal	300	1.000	0.780	0.780
		600	1.000	0.980	0.980
		1000	1.000	0.950	0.950
		2000	1.000	1.000	1.000
0.05	heterogeneous	300	0.760	0.670	0.882
		600	1.000	0.950	0.950
		1000	1.000	0.980	0.980
		2000	1.000	1.000	1.000
0.10	heterogeneous	300	0.060	0.020	0.333
		600	0.630	0.280	0.444
		1000	0.980	0.740	0.755
		2000	1.000	0.970	0.970

The path-oracle audit was available for the $e_B = 0.05$ equal and heterogeneous trajectories and the $e_B = 0.10$ heterogeneous trajectory. The $e_B = 0.10$ equal trajectory was not part of this oracle-path audit and is not imputed.

was not monotone in n, although both path coverage and the oracle-NLL gap improved markedly by $n = 2000$.

S7.1. Classic3 path sensitivity

The original E-series analysis used a single 240-point path with relative lower endpoint 10^{-2}. E-CGL moved gradually from the zero-penalty fit, but the adaptive weights caused the first positive E-ACGL point to move from $q = 2000$ to $q = 442$. A two-segment audit that extended the candidate family down to 10^{-6} showed that the dense E-ACGL selection was a candidate-coverage issue rather than evidence that adaptive weighting intrinsically requires a dense support.

To retain one path per method rather than a dataset-specific union, the final analysis uses a method-specific single $L = 240$ path from $10^{-4}\widetilde{\lambda}_{\max}$ to $\widetilde{\lambda}_{\max}$ for each E-series method. The data, dense starts, fixed weights, target-preserving refit, explicit empty support, and BIC rule are unchanged. Absolute penalty values remain method-specific because the path scale contains the fixed weights.

For E-CGL, the final first positive value was 0.010361 and retained all 2,000 coordinates. The selected refit had $q = 1420$, BIC $-37{,}798{,}212$, ARI

0.9761, and NMI 0.9539. For E-ACGL, the final first positive value was 0.196404 and retained 1,964 coordinates. The selected refit had $q = 1438$, BIC −37,798,237, ARI 0.9746, and NMI 0.9515. The earlier two-segment audit selected $q = 1421$ for E-CGL and $q = 1435$ for E-ACGL; those values are retained only as path-sensitivity provenance.

Table S14: Candidate-path sensitivity of the E-series methods in Classic3. The final rows use one 240-point path with relative lower endpoint 10^{-4}; the other rows record the preceding diagnostics.

Method	Candidate rule	Floor ratio	First $\lambda > 0$	First q	Selected λ	Selected q	BIC
E-CGL	historical single path	10^{-2}	1.036103	1947	3.1827	1421	−37,798,211
E-CGL	final single path	10^{-4}	0.010361	2000	3.1827	1420	−37,798,212
E-ACGL	historical single path	10^{-2}	19.6404	442	0	2000	−37,793,072
E-ACGL	final single path	10^{-4}	0.196404	1964	2.0024	1438	−37,798,237

S7.2. Simulation path sensitivity

The standard-path numerical-study outputs were archived before applying the final E-ACGL candidate rule. We screened 5,025 archived E-series replicate–method candidate rows across 26 cells. For each row, the first-positive retention ratio was the support size at the first positive standard-path value divided by its zero-penalty support size. Both methods had 2,600 archived method–replicate groups with positive and BIC-eligible candidates. In 175 E-CGL groups, the support-deduplicated candidate table did not retain a separate zero-penalty representative because that fitted support was represented by a positive tuning value. The retention ratio was therefore computable for 2,425 E-CGL groups and all 2,600 E-ACGL groups. Table S15 shows that the E-CGL standard path moved gradually from zero, whereas the adaptive weights often produced a much larger first E-ACGL support jump. This screening is a candidate-coverage diagnostic; it does not establish that an uncomputed near-zero candidate would have won the refit-and-BIC comparison.

Table S15: First-positive support retention under the recorded standard E-series paths. Entries summarize the ratio of the first-positive support size to the zero-penalty support size across method–replicate rows.

Method	Rows	Minimum	1%	5%	25%	Median	< 0.90	< 0.50
E-CGL	2425	0.917	0.950	0.958	0.990	0.995	0	0
E-ACGL	2600	0.070	0.080	0.080	0.120	0.350	2369	1609

We then refitted both E-series methods on the augmented union in five targeted cells: the heterogeneous $e_B = 0.05$, $n = 1000$ primary cell; the crossed-support $n = 400$ cell; the pure-concentration $n = 1000$ cell; the equal-concentration $e_B = 0.05$, $n = 2000$ cell; and the weak-beta-min $n = 1000$ cell. Replicates 1–3 were used in each cell for both methods, giving 30 paired fits. All fits were valid. E-CGL selected the standard segment in all 15 repetitions. E-ACGL selected a candidate generated by the added segment in three repetitions, but each had exactly the same support as the archived selection. Across the 30 comparisons, selected q, support, $F_{1,\eta}$, and ARI were unchanged; the largest absolute BIC difference was 4.89×10^{-4}. The targeted audit detected no change in the checked fits.

As a historical sensitivity analysis, E-ACGL was also rerun under the two-segment rule in all 12 primary cells and five stress cells. The exact selected support was unchanged in all 1,200 primary repetitions and in four of the five stress cells. In the weak beta-min cell, an added candidate changed the exact support in 19 repetitions: mean selected size changed from 15.80 to 15.99, $F_{1,\eta}$ from 0.994 to 0.998, exact recovery from 0.81 to 0.93, and ARI from 0.869 to 0.870. Table S16 records this audit without defining the final estimator or implying superiority of adaptive weighting.

Table S16: Historical two-segment sensitivity of E-ACGL to near-zero path augmentation. The 12 primary and five stress cells each contain 100 paired replications; these results are provenance for path development and do not define the final single-path estimator.

Panel A: selected candidate source and support changes.

Cell	Standard	Near-zero	Empty	Support changed
$e_B = 0.025$, equal, $n = 300$	100	0	0	0
$e_B = 0.025$, hetero., $n = 300$	100	0	0	0
$e_B = 0.05$, equal, $n = 300$	100	0	0	0
$e_B = 0.05$, hetero., $n = 300$	100	0	0	0
$e_B = 0.10$, equal, $n = 300$	100	0	0	0
$e_B = 0.10$, hetero., $n = 300$	100	0	0	0
$e_B = 0.025$, equal, $n = 1000$	100	0	0	0
$e_B = 0.025$, hetero., $n = 1000$	99	1	0	0
$e_B = 0.05$, equal, $n = 1000$	100	0	0	0
$e_B = 0.05$, hetero., $n = 1000$	100	0	0	0
$e_B = 0.10$, equal, $n = 1000$	100	0	0	0
$e_B = 0.10$, hetero., $n = 1000$	100	0	0	0
Moderately dense, $n = 300$	99	0	1	0
Moderately dense, $n = 1000$	100	0	0	0
Strongly dense, $n = 1000$	100	0	0	0
High-dimensional, $n = 300, d = 500$	100	0	0	0
Weak beta-min, $n = 1000$	27	73	0	19

Table S16 (continued).

Panel B: paired mean performance, standard/augmented.

Cell	$F_{1,\eta}$	Exact recovery	ARI
$e_B = 0.025$, equal, $n = 300$	0.996/0.996	0.86/0.86	0.927/0.927
$e_B = 0.025$, hetero., $n = 300$	0.996/0.996	0.88/0.88	0.925/0.925
$e_B = 0.05$, equal, $n = 300$	0.993/0.993	0.79/0.79	0.856/0.856
$e_B = 0.05$, hetero., $n = 300$	0.995/0.995	0.85/0.85	0.858/0.858
$e_B = 0.10$, equal, $n = 300$	0.987/0.987	0.77/0.77	0.721/0.721
$e_B = 0.10$, hetero., $n = 300$	0.948/0.948	0.22/0.22	0.702/0.702
$e_B = 0.025$, equal, $n = 1000$	0.999/0.999	0.98/0.98	0.933/0.933
$e_B = 0.025$, hetero., $n = 1000$	0.999/0.999	0.98/0.98	0.928/0.928
$e_B = 0.05$, equal, $n = 1000$	0.998/0.998	0.95/0.95	0.867/0.867
$e_B = 0.05$, hetero., $n = 1000$	0.999/0.999	0.98/0.98	0.869/0.869
$e_B = 0.10$, equal, $n = 1000$	0.999/0.999	0.98/0.98	0.747/0.747
$e_B = 0.10$, hetero., $n = 1000$	1.000/1.000	0.99/0.99	0.753/0.753
Moderately dense, $n = 300$	0.789/0.789	0.00/0.00	0.538/0.538
Moderately dense, $n = 1000$	0.913/0.913	0.00/0.00	0.679/0.679
Strongly dense, $n = 1000$	0.878/0.878	0.00/0.00	0.613/0.613
High-dimensional, $n = 300, d = 500$	0.889/0.889	0.00/0.00	0.767/0.767
Weak beta-min, $n = 1000$	0.994/0.998	0.81/0.93	0.869/0.870

Notes: ‘standard’, ‘near-zero’, and ‘empty’ count the support family that supplied the selected refit. A zero-dimensional winner is counted as the explicit empty-support candidate after support-level deduplication. ‘Support changed’ compares the exact coordinate keys; the same 19 rows also changed selected q. Panel B reports standard/augmented means. The archived standard-path files remain unchanged.

The historical two-segment elapsed times include both segments, support refits, and selection after reuse of the paired dense initialization. They are descriptive run records and are not compared with the end-to-end matched-runtime panel.

Both E-series methods were subsequently rerun on the final single 10^{-4}-floor path in all 12 primary cells, three estimand-separation cells, four $e_B = 0.05$ sample-size cells, and five stress cells. All 4,800 selected refits under the support constraints were finite and converged. Their penalized source fits were audited separately for convergence. The numerical tables in this Supplementary Material use these final single-path results; the preceding standard and two-segment results are retained only in the path-audit tables.

The final hard-trajectory extension added the remaining heterogeneous $e_B = 0.10$ sample-size fits. Across all recorded final-path cells, E-CGL has 2,800 valid fits in 28 cells and E-ACGL has 2,600 valid fits in 26 cells; no fit was nonfinite or nonconverged. The unequal totals arise because the complete hard trajectory was run only for E-CGL.

S8. Ambient-dimension sensitivity

The dimension experiment retains $q_C = 4$ and $q_D = 16$ while varying the number of noise coordinates. Table S17 pools equal and heterogeneous concentration cells and uses target $e_B = 5\%$. At $d \in \{100, 500\}$, each concentration pattern has 50 repetitions, giving 100 after pooling; at $d = 200$, each pattern has 100 repetitions, giving 200 after pooling. These archived sensitivity fits use 120-point standard paths.

Table S17: Ambient-dimension sensitivity of E-CGL and E-ACGL. Entries are means pooled across equal and heterogeneous concentration patterns under the archived 120-point standard-path protocol.

n	d	Method	$\widehat{q}$	$\widehat{q}_D$	$\widehat{q}_N$	$F_{1,\eta}$	ARI
300	100	E-CGL	16.22	16.00	0.21	0.993	0.856
300	100	E-ACGL	16.14	16.00	0.13	0.996	0.856
300	200	E-CGL	16.22	16.00	0.22	0.993	0.859
300	200	E-ACGL	16.09	16.00	0.08	0.997	0.859
300	500	E-CGL	17.96	15.99	1.95	0.972	0.853
300	500	E-ACGL	16.77	15.88	0.89	0.972	0.854
1000	200	E-CGL	16.01	16.00	0.01	1.000	0.865
1000	200	E-ACGL	16.01	16.00	0.01	1.000	0.865
1000	500	E-CGL	16.05	16.00	0.05	0.998	0.867
1000	500	E-ACGL	16.05	16.00	0.05	0.998	0.867

The effect of increasing d was concentrated at $n = 300$; at $n = 1000$, both centered group estimators remained close to the true support at $d = 500$.

S9. Component-number diagnostics

The dense-model K study uses $K \in \{2, \ldots, 8\}$, $K^* = 4$, $n = 1000$, $d = 200$, and target $e_B = 5\%$. Over 50 repetitions, the Akaike information criterion (AIC) (Akaike, 1974) and independent test NLL selected $K = 4$ in every repetition. The Bayesian information criterion (BIC) (Schwarz, 1978) selected $K = 4$ in 39/50 equal-concentration repetitions and 0/50 heterogeneous repetitions. An EBIC-type criterion with index 0.5, the risk inflation criterion (RIC), and its corrected form (RICc) selected $K = 2$ in all repetitions. A 10-repetition five-split holdout diagnostic selected $K = 4$

by minimum validation NLL in 10/10 repetitions under both concentration patterns; the one-standard-error (1-SE) rule selected $K = 4$ in 10/10 equal and 9/10 heterogeneous repetitions.

For a dense K-component fit, the diagnostic used $\mathrm{df}_K = Kd + K - 1$ and the following operational definitions:

$$\begin{aligned}
\mathrm{AIC}(K) &= -2\ell_K + 2\,\mathrm{df}_K, \\
\mathrm{BIC}(K) &= -2\ell_K + \log(n)\,\mathrm{df}_K, \\
\mathrm{RIC}(K) &= -2\ell_K + 2\log(d)\,\mathrm{df}_K, \\
\mathrm{RICc}(K) &= -2\ell_K + 2\{\log(d) + \log\log(d)\}\,\mathrm{df}_K, \\
\text{EBIC-type}_{0.5}(K) &= -2\ell_K + \{\log(n) + \log(d)\}\,\mathrm{df}_K\,.
\end{aligned}$$

The EBIC-type criterion is an operational linear penalty inspired by the large-model-space construction of Chen and Chen (2008); it is not their general combinatorial EBIC expression. The RIC and RICc labels follow Foster and George (1994) and Zhang and Shen (2010), respectively. Here they refer to the recorded high-dimensional penalty forms above and are used only as component-number sensitivity diagnostics.

These findings motivate separation of component-number selection from support selection. Predictive K diagnostics are not presented as a theorem or as a universally valid rule for high-dimensional mixtures.

S10. Rossi-style bridge diagnostic

A five-repetition bridge experiment follows the sparse-prototype setting of Rossi and Barbaro (2022): $n = 200$, $d = 100$, $K = 4$, a target overlap of 5%, and 10% zero entries in the directional means. The mean achieved overlap was 4.70%. Dense vMF, spherical k-means, and the Rossi-style M-L selector had mean ARI 0.824, 0.750, and 0.814, respectively. E-CGL had mean ARI 0.591. Because component-specific zero entries make the coordinate-union posterior-score contrast support dense, this diagnostic targets sparse prototypes rather than the estimand of E-CGL. It is retained as a literature bridge and not as primary evidence for posterior-score contrast support recovery.

S11. Directional companion implementation

The centered mean-direction group lasso (M-CGL) is the directional companion. Let $M \in \mathbb{R}^{K\times d}$ have row k equal to $\mu_k^\top$ and let $H_K = I_K - K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$.

The directional-contrast companion solves

$$\max_{\substack{\pi\in[0,1]^K,\ \mathbf{1}_K^\top\pi=1;\\ 0\le\kappa_k\le\kappa_{\max},\quad \|\mu_k\|_2=1,\ k=1,\dots,K}} \left\{\ell(\pi,\kappa,M)-\lambda_\mu\sum_{j=1}^{d}\|H_K M_{\cdot j}\|_2\right\}.$$

The variable-splitting implementation uses the alternating direction method of multipliers (ADMM) (Boyd et al., 2011). It introduces an auxiliary matrix $\Xi\in\mathbb{R}^{K\times d}$ subject to $\Xi=H_K M$, alternates product-of-spheres gradient/retraction updates with centered group thresholding and scaled-dual updates, and updates each κ_k by numerical inversion of the vMF mean-resultant relation. The Banerjee approximation initializes and brackets the numerical inversion; it is not used as the final M-CGL concentration estimate.

With fixed M, write

$$\rho_k=\frac{r_k^\top\mu_k}{N_k},\qquad A_d(\kappa_k)=\rho_k.$$

Values $\rho_k\le 0$ give $\widehat{\kappa}_k=0$, and positive values are clipped below one. Here $A_d(0)=0$ by continuous extension. When the clipped value is smaller than $A_d(\kappa_{\max})$, the root is found on $[0,\kappa_{\max}]$; otherwise the cap is returned. The Banerjee initializer (Banerjee et al., 2005) is

$$\kappa_{k,0}=\frac{\rho_k(d-\rho_k^2)}{1-\rho_k^2},$$

after which the numerical root, rather than this approximation, is retained. If $\widehat{\kappa}_k=0$, the associated μ_k is not identified; interpretation of directional-contrast support therefore presumes positive-concentration components. All reported selected M-CGL fits satisfied this condition.

The same approximation has different roles elsewhere in the implementation. The dense initializer and M-L use it as a capped concentration update, whereas the penalized E-CGL block optimizes η_k directly and does not invert A_d; only its pooled path-scale calculation uses the approximation. For that calculation, let

$$\bar{\rho}_\bullet=\frac{\|r_\bullet\|_2}{N_\bullet},\qquad \widetilde{\rho}_\bullet=\min\{\max(\bar{\rho}_\bullet,10^{-10}),1-10^{-8}\},$$

and define the final capped approximation

$$\widehat{\kappa}_\bullet^B = \min\left\{\kappa_{\max}, \max\left\{10^{-10}, \frac{\widetilde{\rho}_\bullet(d - \widetilde{\rho}_\bullet^2)}{1 - \widetilde{\rho}_\bullet^2}\right\}\right\}.$$

The implementation sets

$$\eta^{\text{common}} = \begin{cases} \widehat{\kappa}_\bullet^B r_\bullet / \|r_\bullet\|_2, & \|r_\bullet\|_2 > 10^{-12}, \\ 0, & \|r_\bullet\|_2 \leq 10^{-12}. \end{cases}$$

Thus, in the E-series path-scale calculation, the Banerjee approximation is retained as the final pooled concentration rather than numerically inverted. Consequently, the E-series endpoint is a KKT-motivated scale evaluated at the approximate pooled fit, not the exact KKT threshold of the fully optimized all-common fixed-responsibility subproblem. Let $\varrho > 0$ denote the augmented-Lagrangian penalty parameter. For ADMM iteration q, the recorded residuals are

$$R_{\text{pri}}^{(q)} = H_K M^{(q)} - \Xi^{(q)}, \qquad R_{\text{dual}}^{(q)} = \varrho H_K\{\Xi^{(q)} - \Xi^{(q-1)}\} = \varrho\{\Xi^{(q)} - \Xi^{(q-1)}\}.$$

The second equality holds because the Ξ iterates remain column-centered.

For path construction, let $\widehat{\Theta}_\mu^{\text{D}}$ be the dense free-concentration pilot, let $\mathcal{R}^{\text{D}}$ have row k equal to $(r_k^{\text{D}})^\top$, define $D_{\widehat{\kappa}^{\text{D}}} = \text{diag}(\widehat{\kappa}_1^{\text{D}}, \dots, \widehat{\kappa}_K^{\text{D}})$, and set

$$B^{\text{D}} := D_{\widehat{\kappa}^{\text{D}}} \mathcal{R}^{\text{D}}, \qquad \lambda_{\text{proxy},\mu} := \max_{1 \leq j \leq d} \|(H_K B^{\text{D}})_{\cdot j}\|_2.$$

The recorded M-CGL path is

$$\Lambda_\mu := \{0\} \cup \text{Geom}\left(10^{-4}\lambda_{\text{proxy},\mu}, 1.25\lambda_{\text{proxy},\mu}; 239\right).$$

This quantity is a dense-score scale proxy, not a global KKT threshold for the nonconvex observed mixture objective.

The path implementation allows at most 500 ADMM iterations and uses scaled primal and dual tolerances of 10^{-7}; row-norm error on the product of spheres is recorded separately. These are diagnostic controls for the nonconvex companion, not a claim of general convex-ADMM convergence.

For a selected directional-contrast candidate S, the refit removes the sparsity penalty and enforces

$$j \notin S \quad \Longrightarrow \quad H_K M_{\cdot j} = 0,$$

while retaining component-specific concentrations. If $m = |S|$, the generic nominal dimension is

$$\mathrm{df}_{\mu,\mathrm{nom}}(S) = \begin{cases} d + 2K - 2, & m = 0, \\ d + (K-1)m + K - 1, & m > 0. \end{cases}$$

This is a practical BIC dimension under generic local-rank conditions, not an effective degrees of freedom result. The M-CGL null model has a common direction and free component-specific concentrations; it is not the pooled single-vMF model.

The centered-direction penalty is bounded on the product of unit spheres and therefore cannot prevent concentration divergence by itself. The directional companion is consequently fitted with the same finite cap $\kappa_{\max} = 10^6$ as the E-series procedures. This makes the computational objective well posed; it is not a sparsity penalty or a claim that the unconstrained mixture likelihood is bounded.

Because the sphere constraint makes the split problem nonconvex, accepted objective values, primal and dual residuals, and sphere feasibility are reported as numerical checks. Global convergence is not claimed. The final 100-repetition estimand-separation results are reported once in the main article; earlier path-development pilots are not repeated in this Supplementary Material. Table S18 summarizes the four primary companion cells using the directional target S_μ rather than the E-series target S_η.

Table S18: M-CGL performance against its directional-contrast target in the four prespecified companion cells with $e_B = 0.05$. Entries are means with Monte Carlo standard errors in parentheses over 100 paired repetitions.

n	κ pattern	valid	$\widehat{q}$	TPR_μ	FPR_μ	$F_{1,\mu}$	exact S_μ	MSE_μ	ARI
300	equal	100	16.24 (0.05)	1.000 (0.000)	0.001 (0.000)	0.993 (0.001)	0.780 (0.042)	0.000414 (0.000004)	0.856 (0.001)
300	heterogeneous	100	20.90 (0.40)	0.916 (0.010)	0.014 (0.002)	0.898 (0.006)	0.080 (0.027)	0.000804 (0.000037)	0.815 (0.005)
1000	equal	100	16.04 (0.02)	1.000 (0.000)	0.000 (0.000)	0.999 (0.001)	0.960 (0.020)	0.000128 (0.000001)	0.867 (0.001)
1000	heterogeneous	100	20.04 (0.02)	1.000 (0.000)	0.000 (0.000)	0.999 (0.000)	0.960 (0.020)	0.000142 (0.000002)	0.867 (0.001)

Notes: $\widehat{q} = |\widehat{S}_\mu|$. Selection metrics and MSE_μ are evaluated against the centered mean-direction target S_μ; they are not posterior-score-support metrics. All 400 recorded fits were valid.

Table S19 reports the matched end-to-end runtime diagnostic used to describe the relative computational burden.

The final timing panel was run sequentially in one R process on a Windows 10 Pro workstation with an Intel Core i7-11700K processor (8 physical and 16 logical cores) and 31.9 GiB of memory, using R 4.2.1 and Rcpp

1.0.12. The `OMP_NUM_THREADS` and `MKL_NUM_THREADS` variables were unset, so the underlying BLAS/OpenMP thread count was not explicitly pinned. One-time Rcpp compilation was excluded. The table therefore documents implementation burden on the recorded system, not hardware-independent algorithmic speed.

Table S19: Matched end-to-end runtime in the representative heterogeneous cell ($e_B = 0.05$, $n = 1000$, $d = 200$) under the final method-specific path rules. Times are wall-clock seconds over 100 paired repetitions.

method	valid	dense init.	path/refit/select	total median	total IQR
Spherical k-means	100/100	–	–	0.25	[0.22, 0.28]
Dense shared-κ	100/100	–	–	1.59	[1.47, 1.71]
Dense free-κ_k	100/100	4.54	–	4.54	[3.97, 5.23]
M-L	100/100	4.54	49.52	53.84	[48.57, 59.48]
M-CGL	100/100	4.54	503.05	509.61	[415.62, 597.18]
E-CGL	100/100	4.54	70.14	74.62	[69.14, 79.85]
E-ACGL	100/100	4.54	86.82	91.31	[86.16, 96.25]

Notes: Sparse-method totals add the paired dense free-κ_k initialization time to the recorded path, refit, and selector time. M-L retains its event-driven path. M-CGL uses its final single $L = 240$ dense-score-proxy path, whereas E-CGL and E-ACGL use their final single $L = 240$ KKT-motivated paths; all three have relative floor 10^{-4}. The methods solve different optimization problems, so the table describes computational burden rather than algorithmic speed superiority.

S12. Real-data stability and held-out diagnostics

The main article reports one fit to the full cleaned Classic3 payload. This section retains the earlier five prespecified overlapping splits as descriptive stability and held-out checks. The splitwise panel predates the full-data sparse k-means addition and is not pooled with the full-data estimates.

The Classic3 source counts were reconstructed from `Rcoclust` version 0.1.1. The source was the archived CRAN package, and the data object was `data_classic3.rda`. The archive version and object path are fixed by the preparation script; labels from that object were used only to set the benchmark resolution and evaluate fitted clusters.

The full-data payload was encoded with a maximum sequence length of 512 and truncation enabled. The exact encoding identifiers were

Model: `naver/splade-cocondenser-ensembledistil`

Revision: `49cf4c7b0db5b870a401ddf5e2669993ef3699c7`.

Alphabetic tokens of length at least three with document frequency at least two were eligible; the 2,000 largest unlabeled full-payload variances were retained and rows were normalized to unit Euclidean norm. Near-duplicate candidates required cosine similarity at least 0.90 and shingle Jaccard similarity at least 0.75. The final SHA-256 values were

Vocabulary:

`9d4cb302c002212de31a6d5d8e48ad745b6390dcbb83bfe4260a963e3521c53c`

Payload:

`6f4e4842b6bc9e8f274714d4cbd9704f1daacfbabf339a179037c09be7c096e3`.

The full-data ranking above applies only to the full-payload analysis in the main article. For each repeated-holdout split, variance ranking was recomputed from the training documents only; 2,000 Classic3 coordinates or 500 BBCSport coordinates were retained. The unchanged training-selected mask was applied to the corresponding test documents, and the resulting training and test rows were normalized separately. Thus the splitwise held-out results are inductive with respect to feature ranking and do not use test labels or test-document variances.

All reported NMI values were recomputed from the stored assignments using the geometric normalization in Supplementary Section S2.5. This harmonization did not require model refitting or alter any displayed three-decimal full-data NMI.

The historical payload manifest did not retain exact Python, PyTorch, and Transformers versions; none is inferred here.

S12.1. Sparse k-means restart check

The full-data sparse k-means result was checked without changing its label-blind permutation-gap rule. At the selected interior weight bound 3.869962, the recorded $n_{\text{start}} = 30$ fit had selected $q = 985$, ARI 0.137445, objective 35.63645, and cluster sizes 553, 2,546, and 784. Two independent $n_{\text{start}} = 30$ batches at the same bound both selected $q = 1{,}024$, with ARI 0.137273, objective 35.63655, and cluster sizes 553, 2,545, and 785. The adjacent lower and upper bounds gave ARI 0.136553 and 0.137445, respectively. The low ARI was therefore reproducible across these independent starts and adjacent tuning values under the SPLADE representation. This is a local-optimum and implementation sanity check, not a claim about sparse k-means on other representations.

S12.2. BBCSport data and analysis protocol

BBCSport is the five-class document-clustering benchmark distributed by Greene and Cunningham (2006). The source archive contains 737 sports articles in athletics, cricket, football, rugby, and tennis. Normalized-text screening removed 13 exact duplicates; a subsequent audit removed 20 near-duplicates using cosine similarity at least 0.90 followed by token-shingle Jaccard similarity at least 0.75. The final analysis therefore used 704 documents, with class counts 98, 116, 259, 139, and 92, respectively.

Five prespecified stratified 80/20 splits used seeds 20260730–20260734. Each split contained 561 training and 143 test documents. SPLADE used the same model and revision reported above. Within each split, coordinates were ranked by their variance in the training documents only, the leading 500 were fixed, and the same mask was applied to the test documents before separate rowwise Euclidean normalization. Labels were used to form the stratified splits and to evaluate fitted clusters, but not for coordinate ranking, initialization, tuning, support selection, or parameter estimation. The class resolution was fixed at $K = 5$ to match the five supplied benchmark topics. Raw BBC articles are not redistributed because the source content remains under BBC copyright. Table S20 summarizes clustering, support size, and held-out fit across both datasets, and Table S21 reports the Classic3 E-series historical path sensitivity on the same splits.

Table S20: Five-split clustering, support size, predictive fit, and conditional support stability in the two real-data analyses. Values are means (standard deviations) over five prespecified splits.

Data	Method	ARI	NMI	q	Test NLL	Jaccard
Classic3	Spherical k-means	0.970 (0.007)	0.946 (0.010)	2000.0 (0.0)	– (–)	1.000
Classic3	Dense vMF shared κ	0.970 (0.007)	0.946 (0.010)	2000.0 (0.0)	-4872.89 (2.48)	1.000
Classic3	Dense vMF free κ_k	0.973 (0.006)	0.953 (0.011)	2000.0 (0.0)	-4873.97 (2.35)	1.000
Classic3	M-L	0.970 (0.009)	0.947 (0.013)	1924.6 (7.3)	-4872.24 (2.53)	0.982
Classic3	M-CGL	0.970 (0.007)	0.949 (0.011)	1376.8 (15.1)	-4873.34 (2.36)	0.934
Classic3	E-CGL	0.970 (0.007)	0.947 (0.012)	1341.6 (7.2)	-4873.29 (2.33)	0.933
Classic3	E-ACGL	0.973 (0.009)	0.952 (0.014)	1344.2 (8.8)	-4873.30 (2.36)	0.933
BBCSport	Spherical k-means	0.894 (0.022)	0.897 (0.021)	500.0 (0.0)	– (–)	1.000
BBCSport	Dense vMF shared κ	0.898 (0.024)	0.901 (0.020)	500.0 (0.0)	-922.87 (2.35)	1.000
BBCSport	Dense vMF free κ_k	0.877 (0.048)	0.878 (0.032)	500.0 (0.0)	-922.62 (2.79)	1.000
BBCSport	M-L	0.907 (0.020)	0.907 (0.019)	498.2 (0.4)	-921.61 (2.44)	0.997
BBCSport	M-CGL	0.880 (0.050)	0.882 (0.035)	302.8 (21.7)	-921.03 (2.31)	0.864
BBCSport	E-CGL	0.875 (0.053)	0.874 (0.041)	315.0 (43.4)	-921.28 (2.19)	0.835
BBCSport	E-ACGL	0.877 (0.048)	0.878 (0.032)	285.8 (7.1)	-920.81 (2.76)	0.821

Test NLL is reported per document. Jaccard is computed pairwise after restricting each support to the vocabulary available in both training splits. E-CGL and E-ACGL each use a method-specific single $L = 240$ KKT-motivated path with the same relative grid and lower-endpoint ratio 10^{-4}; M-CGL uses its $L = 240$ dense-score-proxy path. M-L retains its Rossi-style event-driven path.

Table S21: Historical Classic3 E-series candidate-path sensitivity across the five prespecified splits. The two-segment rule fitted and refitted a separate 240-point near-zero segment before combining eligible refits for BIC selection. These rows predate the final single 10^{-4}-floor path summarized in Table S20.

Method	Candidate rule	Selected q	ARI	NMI	NLL/doc	Support Jaccard index
E-CGL	Historical single 10^{-2} path	1343.0 (11.8)	0.970 (0.007)	0.947 (0.012)	-4873.30 (2.33)	0.933
E-CGL	Historical two-segment union	1343.0 (11.8)	0.970 (0.007)	0.947 (0.012)	-4873.30 (2.33)	0.933
E-ACGL	Historical single 10^{-2} path	2000.0 (0.0)	0.973 (0.006)	0.953 (0.011)	-4873.97 (2.35)	1.000
E-ACGL	Historical two-segment union	1341.6 (8.9)	0.973 (0.009)	0.952 (0.014)	-4873.29 (2.36)	0.933

S12.3. Paired sparse–dense comparisons

Table S22 reports paired differences between each sparse method and the dense model having the same concentration structure. M-L is paired with dense shared-κ vMF; M-CGL and the E-series methods are paired with dense free-κ_k vMF. Positive NLL differences indicate worse held-out density.

Table S22: Paired five-split sparse-minus-dense differences.

Data	Method	ΔARI	ΔNMI	ΔNLL	ARI not lower	NLL higher
Classic3	M-L	-0.000	0.000	0.650	4/5	5/5
Classic3	M-CGL	-0.003	-0.004	0.631	2/5	5/5
Classic3	E-CGL	-0.004	-0.006	0.675	1/5	5/5
Classic3	E-ACGL	-0.001	-0.001	0.669	3/5	5/5
BBCSport	M-L	0.009	0.006	1.265	5/5	5/5
BBCSport	M-CGL	0.003	0.004	1.592	5/5	5/5
BBCSport	E-CGL	-0.003	-0.004	1.346	4/5	5/5
BBCSport	E-ACGL	0.000	0.000	1.814	5/5	5/5

M-L is paired with dense shared-κ vMF; M-CGL and the E-series methods are paired with dense free-κ_k vMF. Positive ΔNLL denotes worse held-out density fit.

Because the holdouts overlap, these counts are descriptive. In particular, the table is not a noninferiority analysis. Selected q is also not a common estimand across prototype, directional-contrast, and posterior-score contrast methods.

The all-common candidate was also evaluated directly for the full-data Classic3 E-series fits and for each BBCSport training split. It is a single-vMF candidate with nominal dimension d; no penalized path was rerun for this audit. Table S23 shows that it did not minimize BIC in any of the 12 comparisons.

Table S23: Empty-support sensitivity in the final E-series real-data analyses. $\Delta\mathrm{BIC}_0$ is the all-common BIC minus the selected-support BIC; positive values favor the reported selected support.

Dataset	Analysis	Method	Comparisons	Selected q	Min. $\Delta\mathrm{BIC}_0$	Empty-support wins
Classic3	Final single path	E-CGL	1	1420	403501.62	0
Classic3	Final single path	E-ACGL	1	1438	403527.33	0
BBCSport	Final single-path splits	E-CGL	5	287–391	30781.90	0
BBCSport	Final single-path splits	E-ACGL	5	275–292	30892.39	0

Fig. S1 compares held-out clustering accuracy and method-specific coordinate retention across the five splits.

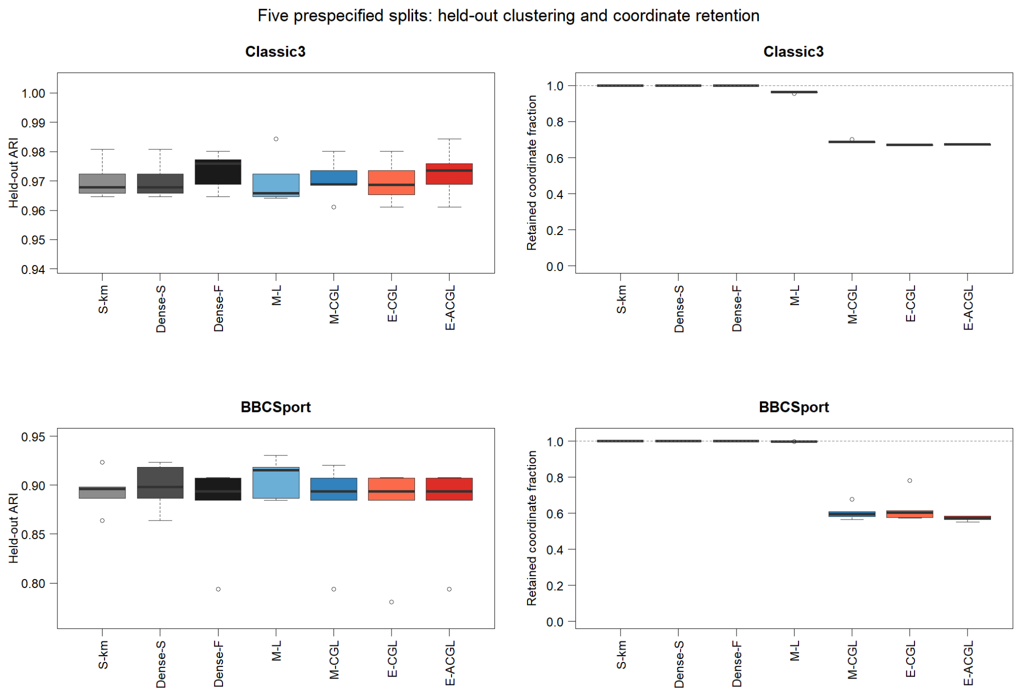

Figure S1: Clustering accuracy and coordinate retention across the five prespecified real-data splits. Coordinate retention is defined relative to each method-specific support target. E-CGL and E-ACGL use the final single 240-point KKT-motivated path in both datasets; M-CGL uses its 240-point dense-score-proxy path, and M-L retains its event-driven path.

S12.4. Classic3 support and token stability

For E-CGL and E-ACGL, the conditional support Jaccard index across the prespecified training vocabularies was 0.933 under the final method-specific single-path rule. Coordinates absent from one of a pair of training vocabularies were excluded from that pair's denominator. The following positive-contrast tokens were available and selected by E-CGL in all five splits. Tables S24 and S25 summarize support overlap and the splitwise selection frequencies of the displayed contrast tokens.

Grouped by the post-fit component labels, the repeated leading positive tokens were *library*, *information*, *librarian*, *libraries*, and *retrieval* for CISI; *flow*, *mach*, *pressure*, *heat*, and *theory* for CRAN; and *tumor*, *inhibitor*, *rat*, *dose*, and *cancer* for MED.

Labels were used only after fitting to name components and display the contrasts. An excluded E-CGL coordinate was constrained to have no centered natural-parameter contrast; its common baseline was retained.

Table S24: Conditional support stability across the five Classic3 training splits.

Method	Mean Jaccard	Range	Available in all	Selected in all	Selected in at least four
Spherical k-means	1.000	1.000–1.000	1791	1791	1907
Dense vMF shared-kappa	1.000	1.000–1.000	1791	1791	1907
Dense vMF free-kappa	1.000	1.000–1.000	1791	1791	1907
M-L	0.982	0.976–0.987	1791	1729	1843
M-CGL	0.934	0.928–0.942	1791	1231	1318
E-CGL	0.933	0.926–0.941	1791	1206	1278
E-ACGL	0.933	0.924–0.943	1791	1208	1283

Pairwise Jaccard indices are computed after restricting both supports to coordinates available in both training vocabularies.

Table S25: Splitwise E-CGL selection frequencies for the 15 selected-contrast tokens displayed in Panel A of Fig. 4.

Token	Available splits	Selected splits	Conditional rate
library	5	5	1.000
information	5	5	1.000
librarian	5	5	1.000
libraries	5	5	1.000
retrieval	5	5	1.000
flow	5	5	1.000
mach	5	5	1.000
pressure	5	5	1.000
theory	5	5	1.000
heat	5	5	1.000
tumor	5	5	1.000
inhibitor	5	5	1.000
rat	5	5	1.000
dose	5	5	1.000
cancer	5	5	1.000

S12.5. BBCSport M-L path sensitivity

The primary M-L path allowed at most 600 path points, including the zero-penalty fit, and imposed a minimum relative penalty increment of 0.02. A sensitivity run allowed at most 1,200 path points, including the zero-penalty fit, and reduced the minimum increment to 0.005. Results are summarized in Table S26.

Table S26: Path-resolution sensitivity of M-L in BBCSport. Path settings and mean diagnostics are reported over the five prespecified splits.

Quantity	Primary path	Denser path
Maximum path points	600	1,200
Minimum relative increment	0.020	0.005
Mean evaluated path rows	331.8	771.2
Mean selected q	498.2	498.4
Mean M-L runtime ratio	1.00	1.76

Selected q differed by at most one coordinate at the split level, and held-out ARI/NMI were unchanged in all five splits. The final evaluated support had BIC at least 19,497.94 larger than the selected support. Both paths nevertheless stopped after a later stronger-penalty fit failed. The near-dense M-L selection is therefore stable to the examined path resolution, but no claim is made about unevaluated candidates beyond the numerical failure.

S12.6. Numerical audit

Table S27 summarizes completion, convergence, boundary, and path-stop diagnostics for the prespecified split analyses.

Table S27: Convergence and boundary diagnostics for the methods included in the final prespecified split-performance tables.

Dataset	Completed	Errors	Nonconverged	Boundary selected	Path fit stops
Classic3	35/35	0	0	0	0
BBCSport	35/35	0	0	0	5

Classic3 M-CGL emitted high-order Bessel precision warnings. Over the selected-fit range, $d = 2{,}000$ and $\kappa \in [607, 818]$, the production Bessel-ratio calculation was compared with a reference based on the second-order uniform

large-order expansion of the modified Bessel function (Olver et al., 2010, Sec. 10.41). With $\nu = d/2 - 1$, $z = \kappa/\nu$, and $p = (1+z^2)^{-1/2}$, the reference retained

$$I_\nu(\nu z) \simeq \frac{\exp(\nu\zeta)}{\sqrt{2\pi\nu}(1+z^2)^{1/4}}\left\{1 + \frac{U_1(p)}{\nu} + \frac{U_2(p)}{\nu^2}\right\},$$

where

$$\zeta = \sqrt{1+z^2} + \log\frac{z}{1+\sqrt{1+z^2}},$$

$$U_1(p) = \frac{3p - 5p^3}{24}, \qquad U_2(p) = \frac{81p^2 - 462p^4 + 385p^6}{1152}.$$

Writing $C_d^{(2)}$ for the vMF normalizer induced by this second-order approximation, the reference ratio was evaluated by the centered finite difference

$$A_d^{\mathrm{ref}}(\kappa) = -\frac{\log C_d^{(2)}(\kappa+h) - \log C_d^{(2)}(\kappa-h)}{2h}, \qquad h = 10^{-5}\max(1,\kappa).$$

The maximum absolute and relative differences were 8.983×10^{-11} and 2.850×10^{-10}, respectively. The observed range passed the prespecified relative tolerance of 10^{-6}; this is not a global approximation-error bound.

A prespecified concentration-guard audit then refit E-CGL and E-ACGL on all five Classic3 and five BBCSport splits under the common `kappa_cap` $= 10^6$ rule. Across the 20 dataset–split–method comparisons, selected supports and held-out assignments agreed in 20/20 cases. The maximum absolute changes in train log-likelihood, test log-likelihood, and the fitted natural parameters were all zero at the stored precision. The largest path and refit ratios $\max_k \widehat{\kappa}_k/$`kappa_cap` were 0.002525 and 0.002537, respectively, and all refits passed the 0.99 eligibility guard. This is a splitwise equivalence audit, not a bound over every possible degeneracy configuration.

S13. CSTR Rossi-style reproduction bridge

The Computer Science Technical Reports (CSTR) collection contains 475 computer-science technical-report abstracts represented by 1,000 binary word coordinates and four reference subject areas, following the benchmark used by Rossi and Barbaro (2022). The implementation reproduced the published clustering levels for the dense shared-κ mixture and the Rossi M-L selector, as summarized in Table S28.

Table S28: CSTR comparison with the published Rossi-style analysis. Implementation results are mean (SD) over 50 runs.

Method	Published ARI	Implementation ARI
Dense vMF, shared κ	0.804	0.8023 (0.0087)
Rossi M-L, BIC	0.808	0.8083 (0.0079)

The earlier centered-η CSTR diagnostic used BIC before refit and is not combined numerically with the prespecified BIC-after-refit analysis. CSTR is used only to check comparability of the Rossi-style implementation.

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